Endogenous bank risks*

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Abstract

According to the empirical literature, bank distress is preceded by credit expansions and accompanied by the joint deterioration of liquidity, credit quality, and solvency, yet existing models treat these risks as separate frictions or exogenous inputs. We build a general equilibrium model in which all three risks arise from a single decision: lending. In our mechanism, banks create deposits when they lend while payment flows across banks generate reserve-settlement needs (liquidity risk), determine the revenues from which borrowers repay (credit risk), and, through loan losses, erode capital (solvency risk). Expanding the balance sheet raises all three exposures, so credit supply is bounded by an internal risk–profitability trade-off. Calibrated to US data, the model also generates interbank premia that rise with borrowing and fall with capital, a bank lending channel that operates without deposit outflows and is stronger for less-capitalized banks, and a risk-taking channel with ex ante identical borrowers.

Keywords: Monetary policy; banks; payments; liquidity risk; credit risk; solvency risk; risk premium; interbank market.
JEL classification codes: E10, E44, E52, G21.

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1 Introduction

Recently, Correia et al. [2026] have shown that bank stress is not one dimensional but rather combines liquidity strains, credit losses, and solvency concerns. In particular, these authors document how bank failures occur after an increasing reliance on expensive and risk-sensitive noncore funding, a rise in non-performing loans, and a gradual deterioration of solvency. Furthermore, asset losses arrive at failing banks after a period of loan expansion. This last conclusion is also supported by a large empirical literature showing that banking crises and financial instability are usually preceded by rapid credit growth [Jorda et al., 2013, Mian et al., 2017, Schularick and Taylor, 2012].

Existing theories of bank fragility have not yet offered a unified account of the joint emergence of credit, liquidity, and solvency risks. Run models in the tradition of Diamond and Dybvig [1983] isolate liquidity: solvent banks fail because depositors coordinate on withdrawing. Macro-financial models following Gertler and Kiyotaki [2010] and Gertler and Karadi [2011] impose agency problems or leverage constraints that operate on the solvency margin, but treat liquidity exposure and credit losses as inputs to the balance sheet rather than as outcomes of the lending decisions that determine solvency. Quantitative models of liquidity management, such as Bianchi and Bigio [2022], treat credit risk as an exogenous parameter instead of tying it to the same balance sheet expansion that drives liquidity exposure. Imbierowicz and Rauch [2014] show empirically that liquidity risk and credit risk each raise banks' default probabilities, and that their interaction matters for default risk, and Pierret [2015] studies the solvency–liquidity nexus.

Our paper fills this gap by developing a framework in which lending decisions and payment flows endogenously link credit, liquidity, and solvency risks. The mechanism works as follows. Banks generate deposits when they extend credit. Once created, these deposits are spent and typically end up at other banks, so the lending bank must transfer reserves to settle the payment. A bank whose depositors' payments exceed the payments it receives must borrow reserves either from the central bank or from the interbank market (*liquidity risk*). The payments that flow to the pool of firms a bank serves are the revenues from which those firms repay their loans. If too few deposits flow back to borrowers, these agents default on their obligations, and the shortfall is the bank's loss (*credit risk.*). Finally, loan losses reduce capital; when they exceed it, the bank is insolvent (*solvency risk.*).

We also show how bank balance sheet expansion is bounded endogenously through the joint effect of liquidity, credit and solvency risks on expected profitability rather than through any external regulatory constraint or exogenous funding limit associated to a pre-existing stock of savings. As a result, there is an optimal level of credit provision for a bank that maximizes the expected profitability. This feature highlights that when bank risks are generated internally through bank operations, credit supply is endogenously restricted by bank risk exposure rather than by a decrease in the funding available. Bank capital is relevant in this process because it affects the connection between loan provision and bank risks. As a consequence, loan supply is, beyond a point, backward-bending in the lending rate, even though information is symmetric between borrowers and banks.

Apart from connecting endogenously lending with liquidity, credit and solvency risks, our

model is also consistent with several empirical patterns found in the literature and not previously connected theoretically. First, the model produces interbank risk premia that rise with borrowing and fall with capital as reported in Arnould et al. [2022], Aymanns et al. [2016], Carvalho et al. [2022], and Schmitz et al. [2017]. In our model, lenders in the interbank market observe a borrower's reserve position, but not whether it arises from a pure liquidity shock or from a shock that threatens solvency. They therefore charge a premium that increases with the amount borrowed and decreases with the bank's capital, as these two variables signal a higher probability of default.

Second, the model describes a bank lending channel that works through risk, not through deposit outflows. A rise in the policy rate makes liquidity costlier, which lowers the profit of expanding the balance sheet at the margin. Banks lend less, and since lending creates deposits, deposits fall too, together with output and leverage. In our economy, these effects happen with no leverage constraint, no capital regulation, and no deposit outflow forcing banks to retrench. Furthermore, we find that the effect of a contractionary monetary policy is larger for less capitalized banks, as found in Altunbaş et al. [2002], Ehrmann et al. [2001], Gambacorta [2005], Gambacorta and Mistrulli [2004], Kashyap and Stein [1995], Kishan and Opiela [2000], Kishan and Opiela [2006], or van den Heuvel [2002].

Finally, the model also produces a risk-taking channel without risk-shifting. Looser monetary policy raises non-performing loans and the share of insolvent banks, as documented by Borio and Zhu [2012], Jiménez et al. [2014], and Dell'Ariccia et al. [2017]. Here this happens although every borrower is ex ante identical: banks take more risk only because lower funding costs lead them to expand credit.

The remainder of the paper proceeds as follows. Section 2 discusses related literature. Section 3 presents the stylized facts from the existing empirical literature. Section 4 presents the model and equilibrium. Section 5 reports results. Finally, Section 6 concludes.

2 Related Literature

Our paper draws on several strands of the literature. We organize the discussion of this section by the question each of these strands answers.

2.1 Liquidity, credit, and solvency risk

The closest literature to our paper asks how liquidity, credit, and solvency risk are connected. On the theory side, He and Xiong [2012] show how short-term debt links liquidity and credit risk through rollover, and Morris and Shin [2016] use global-games methods to decompose credit risk into solvency and illiquidity components that are jointly determined by the balance sheet. Diamond and Rajan [2005] argue that liquidity and solvency problems interact and can cause each other, which makes the root cause of a crisis hard to identify from observable data. In all of these, the balance sheet, and with it the exposure to each risk, is taken as given. In our model it is an outcome: the same lending decision that creates deposits determines how many reserves the bank must find, how much of its loan book borrowers can repay, and how much capital remains.

The empirical literature documents interactions that our model generates. Imbierowicz and Rauch [2014] find that liquidity risk and credit risk have no economically meaningful relationship at the bank level, but that each raises banks' default probabilities, and that their interaction either aggravates or mitigates default risk depending on the bank's overall level of risk. Pierret [2015] and Schmitz et al. [2019] study the solvency–liquidity nexus in funding costs and stress tests, and Correia et al. [2026] show that failing banks combine rising asset losses, deteriorating solvency, and increasing reliance on risk-sensitive funding after a period of loan expansion. The latter finding is consistent with the broader evidence that credit growth predicts banking crises, [Greenwood et al., 2022, Jorda et al., 2013, Mian et al., 2017, Muller and Verner, 2024, Schularick and Taylor, 2012] and that bank losses typically precede panics [Baron et al., 2025].

We contribute to this literature by providing a mechanism that rationalizes these facts together: lending raises all three exposures at once, which explains why they co-move around credit booms, while a pure liquidity shock, which moves reserve positions independently of borrowers' repayment, keeps the correlation between liquidity and credit risk weak at the bank level, as found by Imbierowicz and Rauch [2014].

2.2 Endogenous risk and financial amplification in macroeconomics

A second literature studies how banks' balance sheets amplify shocks. Run models in the tradition of Diamond and Dybvig [1983] isolate one dimension of fragility, liquidity, and have solvent banks fail through self-fulfilling withdrawals. Macro-financial models following Gertler and Kiyotaki [2010] and Gertler and Karadi [2011] introduce agency problems or leverage constraints that operate on the solvency margin, treating liquidity exposure and credit losses as inputs. Brunnermeier and Sannikov [2014] and He and Krishnamurthy [2013] emphasize endogenous risk arising from feedback between asset prices, leverage, and balance-sheet strength. More recently, Jiang et al. [2024] show how rate-induced asset losses can make banks with low capital and high uninsured leverage vulnerable to self-fulfilling solvency runs.

We share the emphasis on risk created inside the financial system, but the source is different: in our model risk arises from the interaction of payment flows with balance-sheet composition, not from agency problems, market power, or beliefs about other depositors' behavior. The bound on credit supply therefore appears even with symmetric information, which distinguishes our backward-bending supply curve from those generated by information frictions. In any case, our mechanisms is complementary with this literature since our model has no runs and abstracts from interest-rate risk and uninsured depositors.

2.3 Money creation, payments, and bank balance sheets

Our modeling of the banking system follows the description of money creation by central banks and academics [e.g., Bailey, 2020, Brainard, 2021, Bundesbank, 2017, Holmes, 1969, Jordan, 2018, McLeay et al., 2014], according to which banks create deposits when they lend and these deposits then circulate as means of payment. Disyatat [2011] and Jakab and Kumhof [2018] use this description to argue that existing theoretical models have the direction of causation reversed, while Werner [2014] documents this fact with the corresponding balance-sheet entries in banks, and Thakor and Yu [2024] show that most US bank deposits can be attributed

to lending. Bindseil and Senner [2024] explain how deposits are destroyed in the aggregate only through banks' own operations. Thus, a depositor who buys an asset from a nonbank merely moves deposits between banks, not affecting the overall funding of the banking sector. Furthermore, we take from Kiyotaki and Moore [2002] the idea that private paper circulates poorly because it is hard to resell, which gives banks' liabilities a role as means of payment.

We contribute to this literature by introducing a tractable approach that embeds payment flows and deposit creation and destruction into a general equilibrium setting with endogenous bank balance sheets. This allows us to study how payment flows affect bank lending and the monetary transmission. Our model jointly accounts for observed relationships between policy rates, spreads, funding costs, and heterogeneous lending responses. To our knowledge, existing macro models examine these channels separately. General equilibrium approaches such as Bianchi and Bigio [2022] and Di Tella and Kurlat [2021] analyze individual bank risks but do not integrate the credit-liquidity-solvency triad in a single structure as we do. Furthermore, dynamic banking models like Bolton et al. [2025] also underscore the value of deposit flows as a constraint to lending.

2.4 The interbank market and wholesale funding costs

A third group of papers analyzes how the interbank market allocates reserves and prices counterparty risk. Afonso et al. [2019] model the federal funds market and show how the distribution of reserves affects interbank rates; Heider et al. [2015] show how counterparty risk shapes interbank rates and liquidity hoarding; and Ashcraft and Bleakley [2006] show that lenders in the federal funds market discriminate among borrowers according to their opacity. A large empirical literature finds that bank solvency is negatively related to funding costs, and more so for market-sensitive sources of funding such as interbank loans [e.g. Arnould et al., 2022, Aymanns et al., 2016, Carvalho et al., 2022, Schmitz et al., 2017].

In our model the premium that borrowing banks pay is not imposed: it is derived from lenders' inability to distinguish a pure liquidity shortfall from one that signals insolvency, and it therefore increases with the amount borrowed and decreases with capital. This connects the interbank premium to the same lending decision that determines the bank's risks. This connection, to our knowledge, is not done in the models above.

2.5 Monetary transmission through banks

The fifth literature looks at how monetary policy affects bank lending. The early bank lending channel [Bernanke and Blinder, 1988, Kashyap and Stein, 2000] relied on reserve requirements or balance-sheet constraints that limit credit creation, and subsequent work on funding structure, market-based intermediation, and liquidity regulation [Adrian and Shin, 2010, Diamond and Kashyap, 2016, Heider et al., 2015, Van den Heuvel, 2008] has enriched this view. Empirical studies show that banks reduce lending more after a contractionary shock when they are less capitalized or less liquid [Altunbaş et al., 2002, Ehrmann et al., 2001, Gambacorta, 2005, Gambacorta and Mistrulli, 2004, Kashyap and Stein, 1995, Kishan and Opiela, 2000, 2006, van den Heuvel, 2002], and Jimenez et al. [2012] show with loan-application data that this reflects credit supply, not only demand. Our model delivers such a lending channel, and because credit supply

is bounded by expected profitability and not by external limits on deposits, capital, or funding, monetary policy acts by changing the cost of intermediation and the maximum attainable profit, not by tightening an exogenous constraint. Heterogeneity across banks arises from capital through the position of the solvency threshold.

Two further strands connect to our results. The risk-taking channel [Borio and Zhu, 2012, Dell'Ariccia et al., 2017, Jiménez et al., 2014] documents that lower short-term rates are associated with higher risk-taking by banks. In our model this arises without borrowers being heterogeneous or banks reaching for riskier borrowers: lower funding costs make credit expansion profitable, and the expansion itself raises exposure. And the deposits channel of Drechsler et al. [2017] argues that, when the policy rate rises, banks with market power widen deposit spreads and deposits flow out, which causes banks to cut lending. We reproduce the joint movement of deposits and lending, but reverse the direction of causation: deposits fall because banks create fewer of them as they lend less, not because outflows force banks to retrench. The frameworks also differ in that banks in Drechsler et al. [2017] do not fail and have no capital, so they cannot generate differences in responses by capitalization.

2.6 Prudential regulation and stress tests

Finally, our results connect to the design of prudential tools. Liquidity and capital standards have largely been designed separately [Basel Committee on Banking Supervision, 2013], whereas several authors argue for tying them together [Tarullo, 2013] and for stress tests that integrate solvency and liquidity [Bank for International Settlements, 2015, Cont et al., 2020, Gauthier et al., 2014]. Although we do not model regulation, our model supplies a reason, grounded in the lending decision, why such integration could matter: if liquidity needs and solvency distress both stem from lending, then each risk carries information about the other, and tools that address them in isolation may be less effective.

2.7 Summary of the contributions

Relative to existing work, the paper's distinguishing features are that (i) the three risks arise from the same lending decision through the payment system and are not separate frictions or exogenous inputs; (ii) credit supply is bounded endogenously by the resulting risk–profit trade-off, with symmetric information; and (iii) a single mechanism accounts for the interbank premium, the heterogeneity of the lending channel by capital, the risk-taking channel, and the co-movement of deposits and lending after policy changes, with causation from lending to deposits.

3 Stylized Facts

This section documents five empirical regularities from the existing literature. The first one motivates the mechanism of endogenous money creation on which the model of Section 4 is built. The other four facts are patterns in the data that this mechanism reproduces. Together they support the claim made in the Introduction that credit, liquidity, and solvency risks are best understood as joint outcomes of lending.

3.1 Fact 1: Deposits are created by lending and destroyed by banks' own operations

Bank lending creates deposits, not the other way around. In the words of the Bundesbank [2017]:

> "Sight deposits are created when a bank settles transactions with a customer, i.e. it grants a credit, say, or purchases an asset and credits the corresponding amount to the customer's bank account in return. This means that banks can create book money just by making an accounting entry: (...) this refutes a popular misconception that banks act simply as intermediaries at the time of lending – i.e. that banks can only grant credit using funds placed with them previously as deposits by other customers."

This view is shared by central banks and academics [e.g., Bailey, 2020, Brainard, 2021, Disyatat, 2011, Holmes, 1969, Jakab and Kumhof, 2018, Jordan, 2018, McLeay et al., 2014]. Werner [2014] documents the balance-sheet entries generated by the extension and disbursement of a new loan. Quantitatively, Thakor and Yu [2024] find that during the 2001–2020 period, about 92 percent of bank deposits can be attributed to lending.

Similarly, Bindseil and Senner [2024] document how deposits are destroyed. According to these authors, deposits can only disappear in the aggregate through internal operations of banks with a simultaneous elimination of an asset or liability without an offsetting entry at another bank. This could be depositors withdrawing cash, paying for an existing loan, buying an asset already in possession of the bank or exchanging the deposit for another bank liability such as debt or capital. A depositor deciding to re-balance her portfolio by buying a security or any other asset from a nonbank, just transfers her deposits from her bank to the bank of the seller of the asset without modifying the aggregate amount of deposits and the capacity of the banking sector to fund lending.

3.2 Fact 2: Credit expansions precede banking distress

Credit growth is a robust predictor of banking crises and financial instability across countries and centuries [Greenwood et al., 2022, Jorda et al., 2013, Muller and Verner, 2024, Richter et al., 2021, Schularick and Taylor, 2012]. The same holds at the level of the individual bank: Correia et al. [2026] show that total loans of US banks that eventually fail grow significantly in the ten years before failure (Figure 1). Related evidence indicates that bank equity losses typically precede, rather than follow, panics [Baron et al., 2025].

3.3 Fact 3: Distress combines credit, liquidity, and solvency problems, but liquidity and credit risk are only weakly correlated at the bank level

Banking distress involves the joint deterioration of credit quality, liquidity, and solvency. Correia et al. [2026] show that failing banks experience rising asset losses and increasing funding vulnerability in the years before failure (Figure 2): panel (a) shows non-performing loans rising and equity and income falling, and panel (b) shows a shift away from demand and insured deposits toward noncore funding. Liquidity strains and solvency concerns also interact: they can

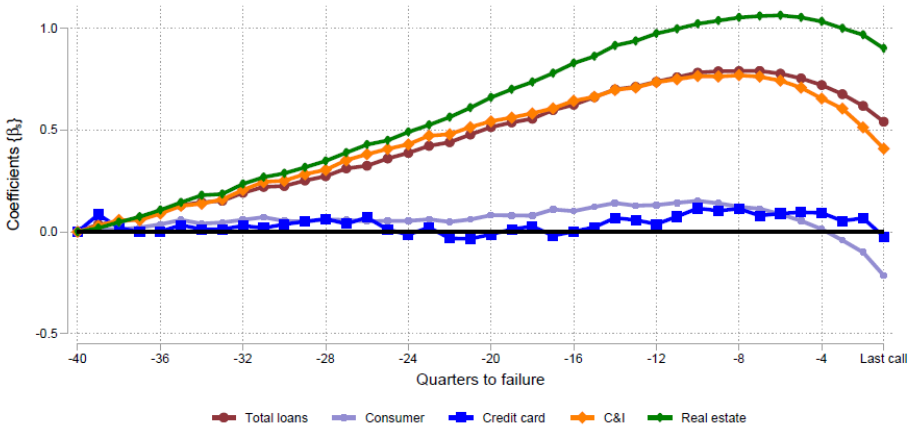
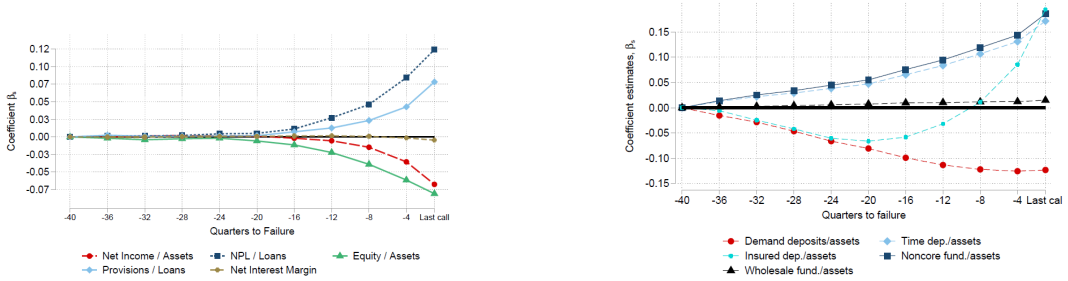


Figure 1: Loan growth at failing banks

Notes: This figure presents the sequence of coefficients from a regression of the following form: $y_{b,t} = \alpha_b + \sum_{j=-10, j\neq 10}^{0} \beta_j \times \mathbf{1} * j = t + \epsilon * b, t$, where $y_{b,t}$ is a type of bank loan. The sample consists of failing US banks in the ten years before failure, based on the post-1959 sample. *Source:* Correia et al. [2026].

cause one another [Diamond and Rajan, 2005], and their interaction affects default probabilities and stress-test outcomes [Pierret, 2015, Schmitz et al., 2019].



(a) Asset losses before failure

(b) Funding vulnerability before failure

Figure 2: Solvency and funding pressures before bank failure

Notes: Panel (a) shows the deterioration of bank asset quality before failure. Panel (b) shows the rise in funding vulnerability before failure. The data cover the period 1959-2024. Together, the two panels illustrate that failing banks experience both solvency problems and liquidity/funding pressures. *Source:* Correia et al. [2026].

A second pattern qualifies the first. Imbierowicz and Rauch [2014] study virtually all US commercial banks over 1998–2010 and find that liquidity risk and credit risk have no economically meaningful relationship, either contemporaneous or lagged: the mean within-bank correlation between their two measures is very close to zero. Yet each risk separately raises the probability of default, and the effect of their interaction depends on the bank's overall level of default risk: it aggravates default risk for banks with a moderate probability of default (about 10 to 30 percent) and mitigates it for banks close to failure (70 to 90 percent). A model of these risks therefore has to deliver two outcomes that look contradictory: strong comovement of the three risks as a bank expands its lending, and a weak correlation between liquidity and credit risk at the bank level.

3.4 Fact 4: The interbank risk premium rises with borrowers' liquidity needs and falls with their solvency

Bank solvency is negatively related to funding costs, and more so for sources of funding that are sensitive to market pressure, such as interbank loans [Arnould et al., 2022, Aymanns et al., 2016, Carvalho et al., 2022, Schmitz et al., 2017]. Liquidity positions matter as well: Ashcraft and Bleakley [2006] find that a bank's liquidity position affects its funding costs in the federal funds market, and Heider et al. [2015] show that counterparty risk is reflected in interbank rates. Quantitatively, Aymanns et al. [2016] estimate that a 1 percent reduction in solvency is associated with an increase of about 4 basis points in wholesale funding costs, and that a 1 percent increase in a bank's liquidity demand raises them by about 7 basis points.

3.5 Fact 5: Monetary policy affects banks differently depending on their balance sheets, risk-taking, and deposits

This fifth fact has three components, each of which corresponds to a result stated in the Introduction.

(a) Heterogeneous lending responses. Banks cut lending more after a contractionary monetary shock when they are less capitalized or less liquid [Altunbaş et al., 2002, Ehrmann et al., 2001, Gambacorta, 2005, Gambacorta and Mistrulli, 2004, Kashyap and Stein, 1995, Kishan and Opiela, 2000, 2006, van den Heuvel, 2002]. Jimenez et al. [2012] use loan-application data to show that this reflects credit supply, not only changes in demand. For example, Gambacorta [2005] estimates for Italian banks a significant fall in lending of 0.825 percent in response to a 1 percent increase in the policy rate, with an average decrease of 0.622 percent for well-capitalized banks and 0.976 percent for poorly capitalized banks.

(b) Risk-taking. Lower short-term rates are associated with greater risk-taking by banks [Borio and Zhu, 2012, Dell'Ariccia et al., 2017, Jiménez et al., 2014].

(c) Deposits and lending move together. Drechsler et al. [2017] document that, after the policy rate rises, banks widen deposit spreads, deposits flow out, and lending falls. Their interpretation is that outflows cause the lending contraction. A joint fall in deposits and lending is, however, also what we would observe if banks lent less for other reasons, since lending creates deposits (Fact 1). The correlation alone does not say which direction of causation dominates.

4 The model

This section builds a general equilibrium model in which the volume of loans a bank decides to supply determines its exposure to liquidity, credit, and solvency risk. The economy is organized in a continuum of nodes, with measure one, indexed by $j \in [0, 1]$. On each of these nodes, there is a continuum of banks, a continuum of firms, and a continuum of workers each with measure one also. Banks supply means of payment. Firms demand labor and produce goods. Workers supply labor and consume. Outside these nodes there is a measure one of investors who save

and invest in firms and banks. Finally, there is a government composed of a central bank and a fiscal authority.

We assume there is specialization in production, consumption and lending. With this assumption, workers and firms on each node are involved in the production of only one good. However, workers on that node consume a different good produced in a different node. This specialization in production and consumption prevents private IOUs issued by firms to circulate as a mean of payment.[1] So, instead, workers and firms rely on private banks, which use their balance sheets to intermediate payments among them. The process works as follows. Whenever a firm demands labor from a worker but does not have enough means of payment to acquire it, a bank creates these means of payment in the form of deposits and lends them to that firm. For that, we assume the firm can only get the loan from a bank in its node. We impose these extreme specialization assumptions to simplify the exposition of the model. Below we show it would be possible to relax them as long as there is no complete diversification of firms and consumers across different goods and of banks across different nodes.

Once created, these deposits are transferred to workers in exchange for labor and circulate in the economy as workers use them to buy goods from other firms. For the original borrower (firm) to pay back the loan she asked for in the first place, the so-created deposits need to get back to her, as this firm sells whatever product she is specialized in to other workers in the economy. Regarding this payment process, we assume that workers' consumption expenditures are randomly distributed across firms. This generates two key effects. First, because workers and firms are distributed across banks, deposits flow unevenly across intermediary institutions. These interbank imbalances are settled using reserves and expose banks to liquidity risk. Second, since some firms receive more revenue than others, there is heterogeneity in loan repayment. Some firms repay in full, while others default. These defaults expose banks to credit risk. When loan losses are sufficiently large, bank capital is eroded, and solvency risk arises. In this way, liquidity, credit, and solvency risks emerge endogenously from banks' lending decisions.

Finally, there is a public sector composed of a central bank and a fiscal authority. The central bank is in charge of backing up this private monetary system through the production of outside money, i.e., cash and reserves. For that, we impose three institutional arrangements we observe in reality. First, the central bank provides commercial banks with reserves which are used to settle accounts between them derived from the asymmetric nature of payments. To manage these reserves, the central bank offers two facilities, a lending facility to provide reserves and a deposit facility to absorb excess reserves. Both facilities imply the same official rate, i^o. Second, we assume deposits to be convertible, on a one-to-one basis, into cash, i.e. the second form of outside money issued by the central bank. Finally, we impose cash to be legal tender, that is, a form of money that courts of law recognize as satisfactory payment for any monetary debt. Thus, in this model, deposits are liquid because, at any time, either can be used directly to make payments or, else, are convertible to an asset that can be used to make payments. Because of this convertibility, deposits earn the same nominal net return as cash, namely, zero. Furthermore, the fiscal authority transfers resources to workers. These transfers are financed by the central bank through reserves.

[1] In words of Kiyotaki and Moore [2002] resaleability of firms' paper is zero in this economy.

4.1 Workers

Workers are hand-to-mouth. At the start of the period a worker supplies labor h^s in the centralized market of her node and receives the wage $W \times h^s$, paid in deposits. Later in the period, after labor is supplied and output has been produced, the worker decides how much to consume. However, she consumes only one of the varieties produced in one of the nodes of the economy. Which variety the worker consumes is random and not known at the time labor is exerted, earlier in the period. Let $c(j)$ be consumption of the variety produced by firms that borrowed from banks in node j, priced at $P(j)$. Preferences are quasi-linear as in Rogerson [1988] or Williamson [2008]:

$$E[\log c(j)] - h^s, \tag{1}$$

and the worker faces the budget constraint

$$P(j)\, c(j) = W h^s + B, \tag{2}$$

where B is a government transfer, paid as a deposit at the worker's bank.[2] Substituting $c(j) = (Wh^s + B)/P(j)$ in (1), the price enters only through an additive term, so the labor supply decision is independent of the price distribution, and the first-order condition gives

$$h^s = 1 - \frac{B}{W}, \qquad c(j) = \frac{W}{P(j)}. \tag{3}$$

Every worker thus spends exactly $Wh^s + B = W$ in nominal terms, whatever the node she buys from.

An important assumption of the model has to do with the distribution of worker's taste shocks over the universe of firms and banks. For that, assume these shocks are not evenly distributed in the economy but rather clustered around firms located in particular nodes and serviced by banks also located there. The result of this clustering is that banks in node $j \in [0, 1]$ will see their firm clients receive a fraction γ of all expenditures in the economy where γ is a random variable with distribution function $\Phi_\gamma(\gamma)$, mean $E(\gamma) = 1$ and support $[0, \gamma_{max}]$.

The idea we want to capture with the shock γ is as follows. Banks in reality are not fully diversified. This lack of diversification arises because each bank provides loans to a different pool of borrowers regarding their geographical location, production sector, or some other characteristic. Because of these differences, revenues from selling their respective goods vary across each pool of borrowers. This variation in revenues affects the ability of those borrowers to pay back the loans they received from their respective banks. This is what generates the credit risk banks face and is what the idiosyncratic bank shock γ is trying to summarize. However, apart from the characterization of γ as a credit shock, we also want to highlight that this shock has a liquidity component as it is associated with payments done with deposits. Because these payments are uneven across banks, they have to be settled with reserves.

[2] The transfer makes labor supply responsive to the wage; see (3).

4.2 Investors

There is a mass one of investors outside the nodes in this economy. They do not consume and only save investing in the net worth of both banks and firms. Each of them starts the period with assets $\mathcal{A}$ split between net worth of firms, $\mathcal{N}$, and net worth of banks, $\mathcal{K}$ for all the nodes in the economy, that is,

$$\mathcal{A} = \mathcal{N} + \mathcal{K}$$

with

$$\mathcal{N} = \int_0^1 N(j)dj,$$

and

$$\mathcal{K} = \int_0^1 K(j)dj.$$

In these expressions, $N(j)$ and $K(j)$ are the initial net worth of each firm and bank in node $j \in [0, 1]$, respectively. Because all firms and banks start identical, they distribute equally these investments across nodes so that $N(j) = N = \mathcal{N}$, and $K(j) = K = \mathcal{K}$.

Denote next period values with a prime. Then, next period wealth of investors is

$$\mathcal{A}' = \mathcal{N}' + \mathcal{K}' - \mathcal{T}$$

where $\mathcal{T}$ are lump sum taxes levied by the fiscal authority. As investors do not consume, they decide how much to invest in firms and banks so that the return on each investment is equalized. That is,

$$\frac{\mathcal{N}'}{\mathcal{N}} = \frac{\mathcal{K}'}{\mathcal{K}}. \tag{4}$$

These returns are computed below.

Because each investor is small relative to the whole pool of investors, they take decisions of firms and banks as given. Next, we describe these decisions.

4.3 Firms

Each firm produces a perishable consumption good of one variety in node $j \in [0, 1]$. The firm starts the period with internal funds, N, decided by investors at the beginning of the period in the form of deposits at a particular bank. The only input needed to produce is labor, h^d, which has to be paid for in advance. Thus, in the event that initial funds N are not enough to cover for the wage bill, $W \times h^d$, the firm needs to borrow the necessary funds from its bank at a rate i^L to be paid for at the beginning of the following period. Thus, the demand for credit by a firm is

$$L^d = max\{0, Wh^d - N\}. \tag{5}$$

Output is $y = (h^d)^\alpha$ with $0 < \alpha < 1$, and is sold competitively at the realized price $P(j)$, with the revenue credited in deposits at the local bank. Firms are risk neutral and have limited liability, so next period's net worth is

$$N(j)' = \max\{0; P(j)(h^d)^\alpha - (1 + i^L)L^d\}. \tag{6}$$

At the time of hiring, the price is uncertain because it depends on γ. As it will be clear below (Section 4.6), the price level depends linearly and positively on the realization of the taste shock γ. Taking this schedule as given, the firm repays its loan if and only if the price exceeds the break-even price

$$P_L = \frac{(1+i^L)L^d}{(h^d)^\alpha}. \tag{7}$$

Let γ_L be the value of the shock at which $P(\gamma) = P_L$. For $\gamma < \gamma_L$ the firm defaults. Expected net worth is then

$$E[N(j)'] = (h^d)^\alpha \left[\int_{\gamma_L}^{\gamma_{max}} P(\gamma)\Phi_\gamma(d\gamma) - P_L[1-\Phi_\gamma(\gamma_L)]\right]. \tag{8}$$

The firm chooses h^d to maximize (8), taking N, W, i^L, and the price schedule $P(\gamma)$ as given. Notice hiring more labor will generate higher output which produces more expected revenues increasing expected profits. However, at the same time, hiring more workers implies a larger demand for credit, L^d. Thus, the threshold price for which the firm starts making profits, P_L, also increases. This effect reduces expected profits. The first-order condition is

$$\frac{\alpha}{1-\Phi_\gamma(\gamma_L)} \int_{\gamma_L}^{\gamma_{max}} P(\gamma)\Phi_\gamma(d\gamma) = (1+i^L)W(h^d)^{1-\alpha}. \tag{9}$$

Given the market wage, W and initial funds, N, labor demand implies a demand for credit as in (5).

4.4 Banks

4.4.1 Initial state and choices

Banks are not diversified: each bank serves the pool of firms of its node. Banks start the period with a balance sheet derived from decisions taken by investors:

$$A = N + K, \tag{10}$$

with assets A in reserves and liabilities consisting of capital K and the deposits N of the firms it serves. The fiscal authority first pays B to workers as deposits, settled in reserves, so the bank holds reserves $A + B$. The bank then decides:

1. how much to borrow from the central bank's lending facility, $O > 0$, or to deposit at its deposit facility, $-(A+B) \leq O < 0$, at the rate i^O;[3] and

2. the volume of credit L to supply to the firms of its node.

Banks do not need existing funds to lend: the defining characteristic of a depository institution is the ability to create deposits when it extends credit. After lending, the balance sheet is

$$(O + B + A) + L = O + D + K, \qquad D \equiv N + L + B, \tag{11}$$

[3]Borrowing from the central bank is senior to the rest of the bank's liabilities, but we assume it is unsecured.

where D are the bank's deposits, all held by workers: $N + L$ paid by firms as wages and B transferred by the government. With all this, the bank has now two types of assets, namely, reserves, $O + B + A$, and loans provided to firms, L. On the other hand, the bank now holds three types of liabilities. The borrowing from the central bank, O, deposits owned by workers, D, and capital, K.

4.4.2 Payment flows and reserve positions

As mentioned above, workers spend all their deposits D to pay for their consumption. But, since expenditures by workers cover all firms in the economy and because this bank is a point in a continuum of banks, these payments imply an outflow of funds for the bank. Let $\mathbb{D}$ be the aggregate flow of payments by workers for consumption goods in the whole economy. We have assumed that a random fraction γ of these payments are channeled to the pool of firms serviced by each bank on each node in the economy. Thus, the net flow for a bank due to the shock γ is $F_\gamma = \gamma\mathbb{D} - D$. In addition to this flow, a pure liquidity shock ϵ moves $F_\epsilon = \epsilon\mathbb{D}$ across banks. As with the shock γ, the shock ϵ is identically and independently distributed across nodes with distribution $\Phi_\epsilon(\epsilon)$, mean $E(\epsilon) = 0$, and support $\epsilon \in [\epsilon_{min}, \epsilon_{max}]$ with $\epsilon_{min} < 0 < \epsilon_{max}$. Each firm ends the period with $\gamma\mathbb{D}$ in deposits, of which only $(\gamma + \epsilon)\mathbb{D}$ remain at the bank that made the loan; the rest is scattered across other banks.[4] The total flow of funds is

$$F = F_\gamma + F_\epsilon = (\gamma + \epsilon)\mathbb{D} - D, \tag{12}$$

a net transfer of deposits and reserves between the bank and the rest of the banking system. The bank's reserve position at the end of the payment stage is

$$R = O + B + A + F = O + B + A + (\gamma + \epsilon)\mathbb{D} - D. \tag{13}$$

Reserves are needed because, to make the payments F, the bank must transfer an asset together with the deposits, and loans are not accepted by other banks as a means of settlement (for instance, because transferring ownership of information-sensitive assets is costly). Central banks provide this generally accepted asset: reserves.

4.4.3 The interbank market

A bank cannot end the stage with a negative reserve position. In the interbank market, banks with a deficit ($R < 0$) borrow from banks with a surplus ($R > 0$). Lenders earn the interbank rate i^R; because a borrower may become insolvent, it pays the premium $s(R)$ on top. This premium is computed below. Using $A = N + K$ and $D = N + L + B$, a bank is a borrower if

$$\gamma + \epsilon < \gamma_R \equiv \frac{D - O - B - A}{\mathbb{D}} = \frac{L - K - O}{\mathbb{D}}. \tag{14}$$

[4]The shock ϵ can be interpreted as payment delays or trade credit between firms, or as firms holding temporary deposit accounts at other banks to manage payments.

Producing more deposits through lending thus has a liquidity cost: it raises γ_R and with it the probability of having to borrow, while holding more reserves, through a lower O, reduces it.[5] Equation (14) also shows that γ_R falls with the bank's own capital K, because capital funds reserves.

4.4.4 Settlement, solvency, and resolution

At the start of the next period, positions are settled. Interest on the central bank facilities is paid in reserves, and the temporary shock F_ϵ is reversed, as firms collect their deposits to repay loans. Each firm then holds deposits $\gamma\mathbb{D}$ and owes $(1+i^L)L$, so firms repay if

$$\gamma \geq \frac{(1+i^L)L}{\mathbb{D}} \equiv \gamma_L. \tag{15}$$

In a symmetric equilibrium, the non-performing loan (NPL) rate is $\Phi_\gamma(\gamma_L)$.[6] If $\gamma \geq \gamma_L$, loans are repaid and bank capital is $K - i^O O + i^L L$ before interbank payments. If $\gamma < \gamma_L$, loans are written off against firms' deposits and capital is $K - i^O O + \gamma\mathbb{D} - L$.

Finally, interbank trades are settled. Let $\Pi^{IB} \equiv \min\{i^R R;\, [i^R + s(R)]R\}$ be the net interbank income of a solvent bank, namely, $i^R R > 0$ for a lender bank in the interbank market ($R > 0$), and $[i^R + s(R)]R < 0$ for a borrower bank ($R < 0$). Capital next period for a bank with conforming loans, $\gamma \geq \gamma_L$, is then

$$K' = K - i^O O + i^L L + \Pi^{IB} \tag{16}$$

Similarly, capital next period for a bank with defaulting loans, $\gamma < \gamma_L$ would be

$$K' = K - i^O O + \gamma\mathbb{D} - L + \Pi^{IB}. \tag{17}$$

In the Appendix we show that if $i^L L + K - i^O O > 0$, which will happen in equilibrium, and the ϵ shock is bounded below, insolvency will occur only within the set of banks with defaulting loans, $\gamma < \gamma_L$. In such a case, a bank is solvent if (17) is non-negative, that is, if

$$\gamma \geq \frac{L - K + i^O O + \Pi^{IB}}{\mathbb{D}}. \tag{18}$$

Because Π^{IB} is small relative to L and K, expression (18) can be approximated by[7]

$$\gamma \geq \frac{L - K + i^O O}{\mathbb{D}} \equiv \gamma_K. \tag{19}$$

Under the condition $i^L L + K - i^O O > 0$, we have that $\gamma_K < \gamma_L$. Thus, if $\gamma \geq \gamma_K$, the bank is solvent. In such a case, if $\gamma \geq \gamma_L$, firms pay back their loans and keep the deposits $\gamma\mathbb{D} - (1+i^L)L$. Otherwise, if $\gamma_K \leq \gamma < \gamma_L$ firms default on their borrowing and lose their deposits.

[5] Here, we assume by the time the bank realizes it has a reserve deficiency the only margin banks have is to borrow in the interbank market and cannot access the lending facility of the central bank.

[6] For an individual bank, either all loans are repaid or none is.

[7] The reason for this approximation is that Π^{IB} depends both on γ and ϵ complicating computations while adding an insignificant contribution to results.

In the event of bank insolvency, $\gamma < \gamma_K$, resolution works as follows. An insolvent bank is closed by a resolution authority, which is part of the fiscal authority. The central bank is a senior creditor and is repaid in full; interbank creditors recover nothing (a loss given default of 100 percent), and the resolution authority settles any resulting surplus or deficit with the government budget. Shareholders of a closed bank receive zero (limited liability). Notice, at the time of the solvency test (19), the only depositors of the bank are the firms which hold a liability position with the bank. Thus, as mentioned above, if the bank is insolvent because firms do not pay bank their loans, these firms loose their deposits.

We can now understand the role of the liquidity shock ϵ. The shock γ affects the liquidity position of the bank. This is because it determines how much funds enter the bank relative to those funds leaving it. At the same time, it also determines whether the bank is solvent or not as it tells the fraction of loans that are repaid, as summarized by the threshold γ_K. If this was the only shock banks face, their demand for funds in the interbank market would be enough for lenders to know if the borrowing bank is insolvent or not. Thus, the role of ϵ is to detach liquidity positions from solvency issues at the time of trading in the interbank market.

4.4.5 Risk premium in the interbank market

The balance sheet and the reserve position of a borrower are public information, but the realization of individual shocks are not known. A lender who observes the position $\bar{R} < 0$ learns, from (13), only the *sum* of the shocks:

$$\gamma + \epsilon = \frac{\bar{R}}{\mathbb{D}} + \gamma_R \equiv \delta(\bar{R}). \tag{20}$$

The borrower repays when solvent, that is, if and only if $\gamma \geq \gamma_K$, where γ_K is defined in (19). By Bayes' rule, the probability that a borrower with position $\bar{R}$ repays is

$$\Psi(\bar{R}) \equiv \Pr\left[\gamma \geq \gamma_K \mid \gamma + \epsilon = \delta(\bar{R})\right] = \frac{\int_{\gamma_K}^{\gamma_{\max}} \phi_\gamma(x)\, \phi_\epsilon\big(\delta(\bar{R}) - x\big)\, dx}{\int_0^{\gamma_{\max}} \phi_\gamma(x)\, \phi_\epsilon\big(\delta(\bar{R}) - x\big)\, dx}. \tag{21}$$

where ϕ_γ and ϕ_ϵ are the densities of γ and ϵ, respectively. Lenders are risk neutral and diversify across borrowers with the same position, so each loan must yield an expected gross return $1+i^R$. Given the assumed resolution procedures, if a borrower becomes insolvent, the lender recovers nothing while a solvent borrower repays $1 + i^R + s(\bar{R})$. Hence, for the lender to recover $1 + i^R$ it must be the case that $(1 + i^R + s(\bar{R}))\, \Psi(\bar{R}) = 1 + i^R$, or

$$s(\bar{R}) = (1 + i^R)\, \frac{1 - \Psi(\bar{R})}{\Psi(\bar{R})}. \tag{22}$$

Proposition 1: Risk premium. Assume the densities ϕ_γ and ϕ_ϵ are log-concave. For given L, O, and $\mathbb{D}$, the premium $s(\bar{R})$ is (i) increasing in the amount borrowed, $|\bar{R}|$, and (ii) decreasing in the bank's capital K.

The proof is in the Appendix. The intuition is that lenders cannot tell whether a large deficit is due to a pure liquidity shock ϵ or to a low γ that threatens solvency. A larger deficit

shifts the posterior toward the latter, and lower capital moves the solvency threshold γ_K to the right.

4.4.6 Choices by banks

For a particular value of γ, call it $\underline{\gamma}$, define the function

$$G(\underline{\gamma}) = \int_{\underline{\gamma}}^{\gamma_{max}} \gamma \Phi_\gamma(d\gamma) - \underline{\gamma}[1 - \Phi_\gamma(\underline{\gamma})]. \tag{23}$$

Then, as shown in the Appendix, the expected net worth of an individual bank next period equals

$$E(K') = \mathbb{D}\left[G(\gamma_K) - G(\gamma_L)\right] + \Pi(O, L) \tag{24}$$

where

$$\Pi(O, L) = i^R \int_{\gamma_K}^{\gamma_{max}} \int_{\epsilon_{min}}^{\epsilon_{max}} R \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) + \int_{\gamma_K}^{\gamma_{max}} \int_{\epsilon_{min}}^{\gamma_R - \gamma} s(R) R \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) \tag{25}$$

is the net profit of trading in the interbank market. The bank then chooses how much to borrow from, or to deposit to, the central bank, O, and the supply of credit, L, to maximize expected net worth next period (24). Notice, choosing these two variables, together with the idiosyncratic realization of the shocks γ and ϵ, determine the reserve position of the bank, R, as defined in (13). Also, when choosing (O, L), we assume the bank takes the interest rate i^R and the premium schedule $s(R)$ as given.

The first order conditions with respect to O, and L are, respectively,

$$i^O = i^R + \frac{1}{1 - \Phi_\gamma(\gamma_K)} \int_{\gamma_K}^{\gamma_{max}} \int_{\epsilon_{min}}^{\gamma_R - \gamma} s(R) \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma), \tag{26}$$

and

$$(1 + i^L)[1 - \Phi_\gamma(\gamma_L)] = (1 + i^O)[1 - \Phi_\gamma(\gamma_K)]. \tag{27}$$

Expression (26) relates the pricing of funds in the interbank market, as summarized by the rate i^R and the premium $s(R)$, with the cost of reserves as determined by the central bank, i^O. On the other hand, expression (27) defines a loan supply curve. It expresses the amount of loans provided by a bank, L, as a function of the loan rate, i^L, given the policy rate, i^O, and its initial capital, K, as well as aggregate quantities like aggregate deposits, $\mathbb{D}$. The supply of loans equate the expected marginal revenue of lending, expressed by the lending rate, to the marginal expected cost. Since deposits created when lending leave the bank, the relevant cost is associated with the rate to be paid to fund this outflow of funds, i^O. Because of limited liability, both these revenues and costs are multiplied by the probability that firms repay the loan and the probability that the bank remains solvent, respectively.[8]

[8]Expressions (26) and (27) should include a term referring to the value of the option, at the margin, of defaulting on interbank obligations when the bank is closed. The Appendix derives these two expressions and shows, for the calibration in the paper, this missing term is quantitatively very small.

4.4.7 Liquidity, credit, and solvency risks and the lending decision

At this point we see the model has produced three thresholds for the demand shock γ. These are the value separating liquid and illiquid banks, γ_R (conditioned on the realization of the pure liquidity shock ϵ), the value determining whether firms pay back their loans or not, γ_L, and the value below which banks become insolvent, γ_K. Larger values for these thresholds increase the liquidity risk, γ_R, the credit risk, γ_L, and the solvency risk, γ_K, of banks.

Proposition 2: Risk exposures. For a given bank, $\partial\gamma_R/\partial L = \partial\gamma_K/\partial L = 1/\mathbb{D} > 0$ and $\partial\gamma_L/\partial L = (1+i^L)/\mathbb{D} > 0$: a larger loan volume raises all three exposures. Moreover,

$$\gamma_K - \gamma_R = \frac{(1+i^O)O}{\mathbb{D}}, \tag{28}$$

$$\gamma_L - \gamma_R = \frac{i^L L + K + O}{\mathbb{D}}, \tag{29}$$

$$\gamma_L - \gamma_K = \frac{i^L L + K - i^O O}{\mathbb{D}}. \tag{30}$$

In a symmetric equilibrium $O = -(A+B)$ and $\gamma_R = 1$, so that $\gamma_K < \gamma_L < \gamma_R$ if and only if $i^L L < N + B$.

The proof is in the Appendix. By looking at expressions (14), (15), and (19), we can see that, given initial conditions of the bank as measured by initial assets, A, and liabilities, K and N, as well as the population value of deposits, $\mathbb{D}$, and prices, i^L, and i^O, larger loan production, L, jointly increases all of these risks for a single bank. Because these risks affect the expected profits of the bank, as summarized in (24), they are the endogenous limits to the expansion of the bank's balance sheet through the choice of loans and deposits provision. Furthermore, the ordering $\gamma_K < \gamma_L < \gamma_R$ means that relatively bad realizations of γ first make the bank illiquid, worse ones cause firms to default, and only very bad ones render the bank insolvent. This ordering holds because banks are net depositors at the central bank in equilibrium, and because interest on loans, $i^L L$, is small relative to $N + B$; the condition is satisfied at the calibration in Section 5.

Figure 3 plots expressions (14), (15) and (19) in the (γ, ϵ) plane. The decreasing green line is the locus $R = 0$ from (14), with γ_R being its intercept at $\epsilon = 0$: banks below this line are illiquid and borrow, and banks above it lend. The blue vertical line represents (15) separating banks with defaulting loans from banks with conforming loans. Finally, the red vertical line shows (19) and separates solvent and insolvent banks. In this manner, expressions (14), (15) and (19) divide the population of banks in six groups regarding their liquid/illiquid, performing/non-performing loans and solvency/insolvency situation. Each of these six areas in Figure 3 contribute differently towards expected profits of the bank. The area to the left of γ_K contributes $-K$ to expected profits; the area to the right of γ_L and above the locus $R = 0$ contributes the profit on performing loans plus the interbank lending income; and to the right of γ_L but below the locus $R = 0$ the bank earns the loan margin but pays for interbank borrowing. A larger L shifts all three loci to the right, enlarging the areas of illiquidity, default, and insolvency. The bank chooses L by comparing these costs with the revenue from lending.

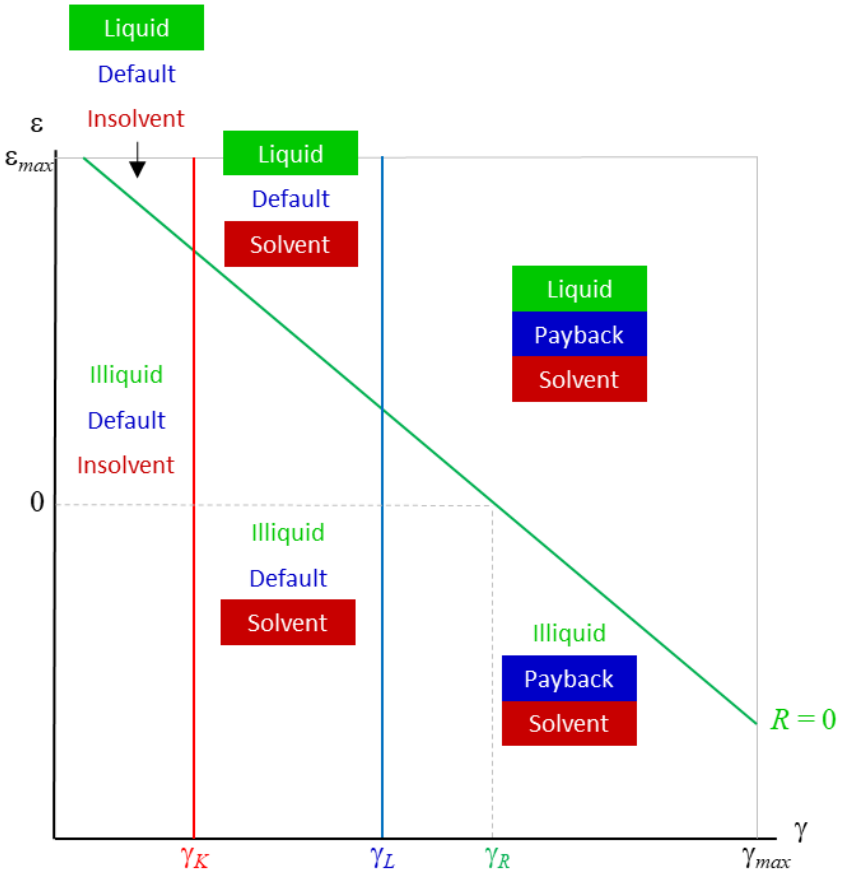


Figure 3: Solvency, credit, and liquidity risks

Notes: The figure summarizes the three endogenous risks generated by the model. For sufficiently low realizations of the shocks, borrowers default and banks are exposed to credit losses. If these losses are large enough to push bank net worth below the solvency threshold, the bank becomes insolvent. Even when the bank remains solvent, it may face liquidity risk if reserve holdings are insufficient to settle net payment outflows.

4.5 Consolidation of accounts and the government

By the law of large numbers, expected values per node equal cross-sectional averages. The settlement of individual bank and firm positions results in a distribution of net worth of banks, $K(j)'$, and firms, $N(j)'$, across all nodes in the economy. At the beginning of each period, after settling individual positions, investors pool the net worth of the distribution of banks and firms. For the conglomerate of firms, assuming all of them have taken the same decisions through the previous period, consolidating their net worth would imply an aggregate net worth for each investor of

$$\mathcal{N}' = \mathbb{D}G(\gamma_L), \tag{31}$$

where the function $G(\gamma)$ was defined in (23). Notice the resulting net worth of the conglomerate of firms is in the form of deposits in all the banks.

Consolidated bank capital is

$$\mathcal{K}' = \mathbb{D}\left[G(\gamma_K) - G(\gamma_L)\right] + \mathcal{P}, \tag{32}$$

where $\mathcal{P} \equiv \Pi(O, L)$ evaluated at the symmetric equilibrium is the net interbank income of the banking system. It is the aggregate counterpart of the interbank term in (17). Because borrowers pay a premium that compensates lenders for their expected losses, $\mathcal{P}$ is non-positive up to terms involving insolvent lenders.

The Appendix shows that, in a symmetric equilibrium,

$$\mathcal{N}' + \mathcal{K}' = A + B - i^O O + \Sigma + \mathcal{P}, \tag{33}$$

where Σ is the expected deficit of insolvent banks, the negative part of (17) and is equal to

$$\Sigma \equiv \mathbb{D}\big[G(\gamma_K) - (1 - \gamma_K)\big] \geq 0.$$

The central bank provides or drains reserves in the aggregate through its facilities at the rate i^O, monetizes the transfers B and the resolution deficits, and the fiscal authority levies the lump-sum tax T on investors, paid in reserves, so as to keep the reserve stock constant. The government budget constraint is therefore

$$T = B - i^O O + \Sigma + \mathcal{P},$$

which, with (33), is equivalent to investors' wealth satisfying $\mathcal{N}' + \mathcal{K}' - T = A$: the tax-and-transfer scheme resets investors' wealth every period, which keeps the model stationary without introducing wealth distributions as state variables. In a symmetric equilibrium all banks choose the same O, and B and T are the same for all agents.

4.6 Equilibrium

Market clearing conditions are as follows. Market clearing for labor means that

$$h^s = h^d = h \tag{34}$$

for all workers and firms where labor supply is given by (3) and labor demand by (9). Thus, output for every firm is $y = h^\alpha$ and market clearing for goods imply

$$\gamma c(j) = h^\alpha \tag{35}$$

for all $j \in [0, 1]$, where γ is the realization of taste shocks for that particular node. Furthermore, given clearing in the labor market, for the loan market to clear it has to be the case that

$$L^d = Wh - N = L, \tag{36}$$

for all banks. Finally, clearing in the interbank market implies

$$\int_0^1 R(j)dj = 0. \tag{37}$$

We define a symmetric equilibrium as follows.

Definition (Symmetric competitive equilibrium). *Given a monetary policy as defined by the official rate, i^o, and a fiscal policy as defined by equal taxes to all investors, $\mathcal{T} = \mathbb{T}$, and equal transfers to all workers, $B = \mathbb{B}$, a symmetric competitive equilibrium is a set of prices $\{P(j)\}$, for all $j \in [0,1]$, interest rates, i^L and i^R, risk premia, $s(R)$, aggregate allocations, $\mathbb{O}$, $\mathbb{N}$ and $\mathbb{K}$, as well as individual allocations $\{c(j), h, y, K, N, O, L, R(j)\}$, for all $j \in [0,1]$ such that, (i) given prices and risk premia, individual allocations solve the problems of all workers, investors, firms and banks, (ii) markets for goods, labor, loans and reserves clear, (iii) individual choices coincide with aggregate ones, and (iv) the government budget constraint is satisfied.*

4.7 Discussion

In this economy, each worker contributes to the production of one good but consumes from a different firm in a different node randomly assigned to her. Once all decisions are taken, consumption expenditures by workers are unevenly distributed across nodes. This crucial assumption incorporates the idea that banks are not fully diversified in their credit risks. Because production is predetermined, a larger flow of payments generates larger revenues for the corresponding firms which, in turn, will be able to pay back the loans they asked for before. On top of these payment shocks, labeled γ in the model, we incorporate other, pure liquidity shock, ϵ, to detach the liquidity position of banks from their solvency situation.

In this model, there are five important frictions we assume. First, because of the nature of production and preferences, private IOUs issued by any firm have no market value for other firms in different nodes. This is because the good is perishable and because of the continuum of workers and firms, finding the right person to trade these IOUs with would be impossible. In other words, there is no double coincidence of wants. Banks can represent a cheap and implementable alternative to these markets in firms' IOUs. In this way, banks are in charge of producing the units of account, i.e. deposits, used to keep track of trades among workers and firms. Similarly, the second friction has to do with banks not having assets generally acceptable by other banks at the time payment flows occur. For that problem, the central bank acts as a bank of banks and issue reserves tracking down the net payment flows between banks. What this model exploits is the connection between the production of inside money when producing loans and the risks associated with providing credit to the economy. Furthermore, monetary policy affects this connection by fixing the cost of the outside money needed to settle the corresponding payment flows. The third friction assumes that banks cannot infer credit risks from deposit flows. In other words, they observe the sum $\epsilon + \gamma$ and not each component separately.

The fourth friction has to do with restricting firms to borrow from only one bank (or a collection of banks) in its very same node. Finally, the fifth friction imposes that neither workers, firms or banks can insure against the idiosyncratic payment shocks occurring at the node level. In this sense, we could generalize the model by allowing firms to borrow from a subset of banks in different nodes or include the possibility of some insurance. We could also complicate the model by adding the possibility that firms and bankers consume too. Or by allowing workers to borrow or save. However, as long as there is no full insurance against payment shocks, agents will need financial liquidity to settle trade positions, there would be payment flows with liquidity, credit, and solvency risks involved, and the results found in this

paper would still apply,

Ultimately, the risk firms face in this economy is the inability to attract enough revenue to pay back the loan they asked for. If severe enough, this credit risk can generate insolvency in the banks that created those means of payments in the first place. Notice this is not an issue of asymmetric information between borrowers and lenders but the inability of the existing financial intermediaries to fully insure against the payment risks in the economy. This is the crucial friction behind the results produced with this model.

5 Results

This section takes the theory to the data. We proceed in four steps. First, we calibrate the model to quarterly U.S. data for 1986–2021, disciplining the two shock distributions and the fiscal transfer with three targets, namely, the average TED spread, average bank leverage, and the average NPL ratio (Section 5.1). Second, we show that the model generates the co-movement of liquidity, credit, and solvency risk together with a weak cross-sectional correlation between liquidity and credit risk (Section 5.2). Third, we show that the model generates, as untargeted implications of the same mechanism, a risk premium in the interbank market that rises with the amount a bank borrows and falls with its solvency, matching the documented solvency–funding-cost nexus (Section 5.3). Fourth, we use the calibrated model to characterize the bank lending channel in the long run across steady states indexed by the policy rate (Section 5.4). Finally, we show that the lending response to a monetary contraction is larger for less-capitalized banks, in line with the cross-sectional evidence on the lending channel, and we trace this heterogeneity to the position of the insolvency threshold in the space of payment shocks (Section 5.5). Except for the three targeted moments in the calibration, all other results are untargeted implications of the same mechanism.

5.1 Calibration

The equilibrium of the model is neutral with respect to the value of the initial liquid bank assets $\mathbb{A}$. For this reason, we normalize this variable to be $\mathbb{A} = 1$. Then, the model contains only one parameter, two distributions, and two policy variables to calibrate. These are the span of control, α, the distributions of shocks, $\Phi_\gamma(\gamma)$ and $\Phi_\epsilon(\epsilon)$, as well as the policy rate i^O and the transfers from the government, B.

Among the parameters of the model, we set the value of the span of control to $\alpha = 0.85$ as in Atkeson and Kehoe [2005]. The rest of parameters are calibrated using quarterly data for the US between the first quarter of 1986 until the fourth quarter of 2021. This is the period for which we could retrieve data for most of the variables considered. In the baseline, we fix the value for the policy rate, i^O, to be 3 percent which corresponds to the average of the federal funds rate (FFR) over that time period. Furthermore, we assume normal distribution functions for γ and ϵ. The distribution of γ is truncated at zero, which is innocuous at the standard deviation reported below (zero is more than four standard deviations from the mean); the distribution of ϵ is truncated symmetrically around zero, consistent with the support $[\epsilon_{\min}, \epsilon_{\max}]$ with $\epsilon_{\min} < 0 < \epsilon_{\max}$ defined in Section 4. We fix the means of these distributions to be $\mu_\gamma = 1$

and $\mu_\epsilon = 0$ so that, on average, no funds enter or leave the banks. The standard deviations, σ_γ and σ_ϵ, and the transfer from the government, B, are calibrated to match three targets. One of the targets is the average risk premium in the interbank market as measured by the TED spread, that is, the difference between the 3-month LIBOR and the 3-month T-Bill. In the model this number is computed as in the data, namely a simple average of rates payed by borrowing banks with the highest and lowest quartiles removed. We also use quarterly data from the Flow of funds accounts of the US to match two ratios the balance sheet of banks. These two ratios are leverage, L/K, and the NPL ratio, denoted in the model by the distribution $\Phi_\gamma(\gamma)$ evaluated at the firm default threshold, γ_L.[9]

The matching of the balance sheet of banks in the model and in the data deserves an important comment. Obviously, the balance sheet of banks in the data is a more complex object than in the model. In particular, loans in the model mature in one period while in the data there is a whole distribution of loan maturities. Similarly, the bulk of deposits in the model are produced in one period and disappear the following one as they are used to pay for the existing loans. In reality, a large stock of deposits, saved by their owners and not used for transactions, funds a large fraction of the stock of long term loans. In this sense, the model could be interpreted in two ways. The first one, more restrictive, suggests the model is designed to reproduce the dynamics of short term borrowing such as credit card or working capital loans. The second one, more general and preferred to us, is that the model concentrates on the flow of assets and liabilities of banks rather than stocks. In this sense, every period, there is production of loans and there is a fraction of the stock that matures. At the same time, new loans add to the existing stock of deposits, a fraction of which is used to make payments generating revenue for borrowers to pay back their loans. The model is designed to analyze these flows. Of course, the dynamics of the stocks could also be important. But the message of the paper would still go through, at least qualitatively. Larger stocks of loans would generate a larger flow of loan repayments which still have to be covered with revenues obtained by borrowers from the flow of payments using a fraction of the stock of existing deposits.

Table 1 presents the calibration exercise. As the table shows, the calibrated value for the parameters are $\sigma_\gamma = 0.22$, $\sigma_\epsilon = 0.31$, and $B = 0.5$. With these values we closely match the data targets on the TED spread, bank leverage and the NPL ratio. Below these targets we also include comparisons of other variables we have data for. First, the model falls a little bit short of approximating the fraction of insolvent banks. In the model, this ratio is measured by the distribution $\Phi_\gamma(\gamma)$ evaluated at the bank insolvency threshold, γ_K. Furthermore, the model also closely matches the size of the nonfinancial sector measured by the endogenous variable N, as a fraction of A. It also produces a higher deposit to loan ratio, D/L, than the one found in the data.

Figure 4 shows loan supply and demand for the calibrated values of parameters and the equilibrium values for other variables such as the net worth of firms, N, or bank capital, K.[10] We can see how supply is elastic at the level of the official rate i^O which in this case equals 0.03 (or 3 percent). Also, loan supply is backward bending. Given the capital of the bank, increases

[9]Regarding the limits of the distributions for γ and ϵ, γ_{max}, ϵ_{min} and ϵ_{max}, we set these numbers to values for which probabilities are close to zero.

[10]Please notice lending rates are in the horizontal axis while loan quantities are in the vertical axis.

External parameters					
Param.	Value	Source			
α	0.85	Taken from Atkeson and Kehoe [2005]			
Targeted moments					
Param.	Value	Variable	Meaning	Data	Model
i^O	0.03	i^O	Federal funds rate	0.03	0.03
σ_γ	0.22	L/K	Bank leverage	7.3	7.2
σ_ϵ	0.31	$\int s(R)dj$	TED spread	0.005	0.005
B	0.5	$\Phi_\gamma(\gamma_L)$	NPL ratio	0.021	0.023
Untargeted moments					
		Variable	Meaning	Data	Model
		$\Phi_\gamma(\gamma_K)$	Insolvent banks	0.013	0.006
		N	Size of nonfinancial sector	0.828	0.792
		D/L	Deposits to loans	1.2	1.8

Table 1: Calibration and model fit

in the lending rate i^L raise marginal revenues inducing the bank to lend more. However, more credit provision also increases risks for the bank, as measured by thresholds γ_R, γ_L and γ_K. As this process continues, there is a point where these risks have grown so much that it is not worth it for the bank to increase lending further as a response to larger lending rates. In fact, beyond that point the bank reduces its credit provision for further increases in lending rates. This is the tipping point of the loan supply curve depicted in Figure 4. Notice, unlike models with agency problems, the supply schedule is backward bending despite information being symmetric between borrowers and lenders.[11]

5.2 The endogenous correlation between bank risks

In subsection 4.4.7 we discussed how loan provision by banks generates a joint increase in liquidity, credit and solvency risks. Although in the model all banks start identical and make the same lending choices, it is possible to argue that, if for exogenous reasons a bank decides to expand credit above its peers, it will concentrate all three risks simultaneously. Thus, the model can account for a positive correlation between these three risks driven by changes in lending.

The model also helps to understand why Imbierowicz and Rauch [2014] find no economically meaningful relationship between liquidity risk and credit risk at the bank level, even though each raises the probability of default. They proxy liquidity risk by the short-term liabilities that can be withdrawn on short notice, net of assets that can quickly be turned into cash, relative to total assets, and credit risk by net loan losses relative to the previous year's loan-loss allowance. We build the model counterparts. Since all banks start with the same balance sheet, we measure liquidity risk by the reserve shortfall, $LR = -R$, which is positive when a

[11]The loan-supply schedule is concave because the marginal expected return to raising the lending rate declines: higher i^L increases the repayment/solvency cutoffs, generating a more-than-proportional rise in credit and solvency risk. If banks could recapitalize costlessly within the period, they would partly offset the increase in solvency risk, flattening the supply schedule and delaying the point at which it bends backward. The schedule would nevertheless remain concave, and eventually backward-bending, because higher lending rates still worsen borrower repayment probabilities.

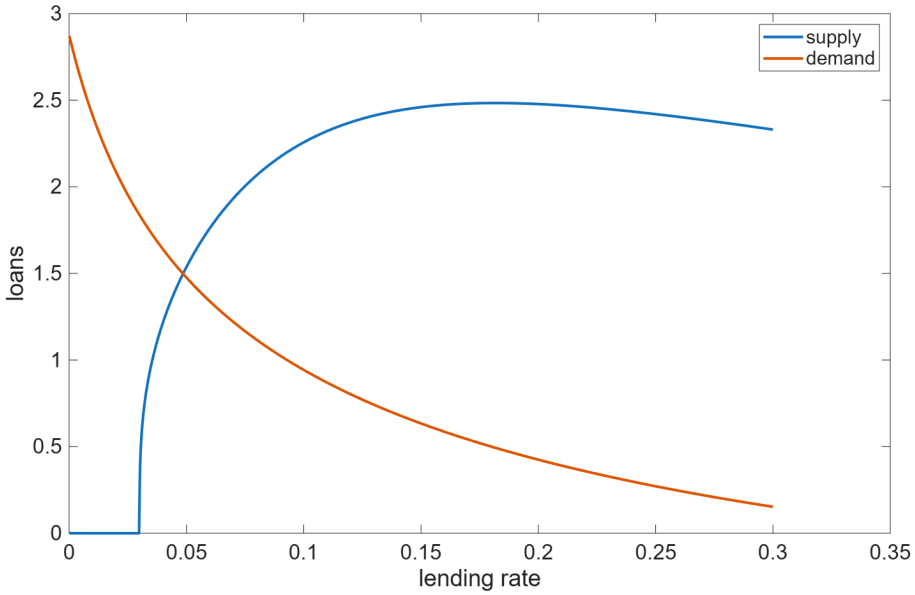


Figure 4: Loan supply and demand

bank must borrow, and credit risk by net charge-offs, $CR = \max\{0, (1 + i^L)L - \gamma\mathbb{D}\}$. Both are computed ex post, once the shocks are realized, which produces a cross-section of banks. The model-implied correlation is $\text{corr}(LR, CR) = 0.10$, an untargeted implication of the calibration, since σ_ϵ was chosen to match the interbank premium.

In the right-hand panel of their Table 5, Imbierowicz and Rauch [2014] report the mean of within-bank correlations between their two measures. For all banks it is 0.006 before the financial crisis and -0.031 during it (significant at 1 percent); by bank size the estimates range from -0.073 to 0.014, and only a few are statistically significant. Because the banks in our model are identical and the shocks are independent across nodes, the correlation across realizations of the shocks coincides with the within-bank correlation over time, so the two are comparable. Both are small: the model's 0.10 is also a weak relation, in line with the authors' conclusion that the two risks have no economically meaningful relationship. The model value is, however, larger in absolute value than the all-bank estimates, and it is positive whereas the crisis-period estimates are negative. We therefore read the result as a qualitative agreement (a weak bank-level correlation despite a common source of risk), not as a match with a point estimate.

Two features of the model contribute to this result. First, net charge-offs are positive only for the small fraction of banks whose firms default, $\Phi_\gamma(\gamma_L) \approx 2$ percent at the calibrated parameters, whereas reserve positions vary across all banks. Second, the pure liquidity shock ϵ moves reserve positions without affecting borrowers' repayment, and has a large volatility relative to that of γ ($\sigma_\epsilon = 0.31$ against $\sigma_\gamma = 0.22$). Neither feature changes the fact that, for any given bank, a larger loan volume raises liquidity, credit, and solvency risk together (Proposition 2).

5.3 Risk premium in the interbank market

Aymanns et al. [2016], Arnould et al. [2022], Carvalho et al. [2022], and Schmitz et al. [2017] present evidence on how solvency ratios are negatively related to bank funding costs. These

authors notice that this negative relationship is stronger for funding sources more sensitive to market pressure, such as interbank lending. For example, Aymanns et al. [2016] estimated that a 1 percent reduction in the level of solvency is associated with a significant increase of 4 basis points in wholesale funding costs. Interestingly, they also note that bank's liquidity position negatively affects funding costs, a result also found in Ashcraft and Bleakley [2006].

Section 4.4.5 shows the determination of the risk premium borrowing banks have to pay in the interbank market, $s(R)$. To understand the intuition behind this computation, it is useful to plot the combination of realizations for the shocks γ and ϵ generating a particular reserve position. Using expression (13), a bank with a given initial state and decisions on endogenous variables would end up with a specific reserve position, call it $\bar{R}$, if the realizations of the shocks fulfill the expression

$$\gamma + \epsilon = \frac{\bar{R} + D - O - A - B}{\mathbb{D}} = \frac{\bar{R}}{\mathbb{D}} + \gamma_R. \tag{38}$$

Figure 5 includes two of such schedules for two different negative levels for the reserve position of a bank, R_{low} and R_{high}, such that $|R_{low}| < |R_{high}|$. Notice the schedule with the largest borrowing, R_{high}, is below the one with banks borrowing less, R_{low}. Furthermore, the section of these schedules in the region to the left of the vertical line associated with the value for γ_K, represent realizations of the shocks inducing the bank to be insolvent and, therefore, will imply a default on the interbank loan.

Grafically, the risk premium is proportional to the density associated with the segment of schedule (38) inside the insolvency area, that is, to the left of the level γ_K. That segment is marked in red in Figure 5. As the amount borrowed gets larger, the schedule moves down and the segment representing insolvency gets larger increasing the risk premium the bank needs to pay. In other words, because banks in the market cannot differentiate between pure liquidity (ϵ) and liquidity/solvency (γ) shocks at the time they reach the interbank market, they assign a higher probability of default to those banks borrowing more wholesale funds and, therefore, a larger risk premium. Similarly, other things equal, a bank with lower capital levels will face a higher threshold for solvency, γ_K. As a consequence, there is pressure for the risk premium, $s(R)$, to be larger too. This is the probability computed in expression (21).[12]

Figure 6 presents the spread over the interbank rate borrowing banks have to pay, $s(R)$, as a function of the amount they borrow, R. All computations are done assuming a policy rate of 3 percent and for the calibrated parameters in Table 1). As banks borrow more, the spread they have to pay over the interbank rate increases up to 80 basis points for very large demands of reserves. We also observe how this relation is nonlinear with spreads being proportionally larger the more the bank borrows as this is a signal of higher probability of insolvency.

5.4 The lending channel redux

In our model, changes in the policy rate i^O engineered by the central bank have effects in the economy. The transmission mechanism works as follows. As the central bank raises the policy rate, commercial banks find that liquidity is costlier. In terms of Figure 3, the cost associated

[12]Of course, endogenous variables will not be equal for two banks with different initial capital levels so the overall effect on the spread will depend on endogenous choices of banks.

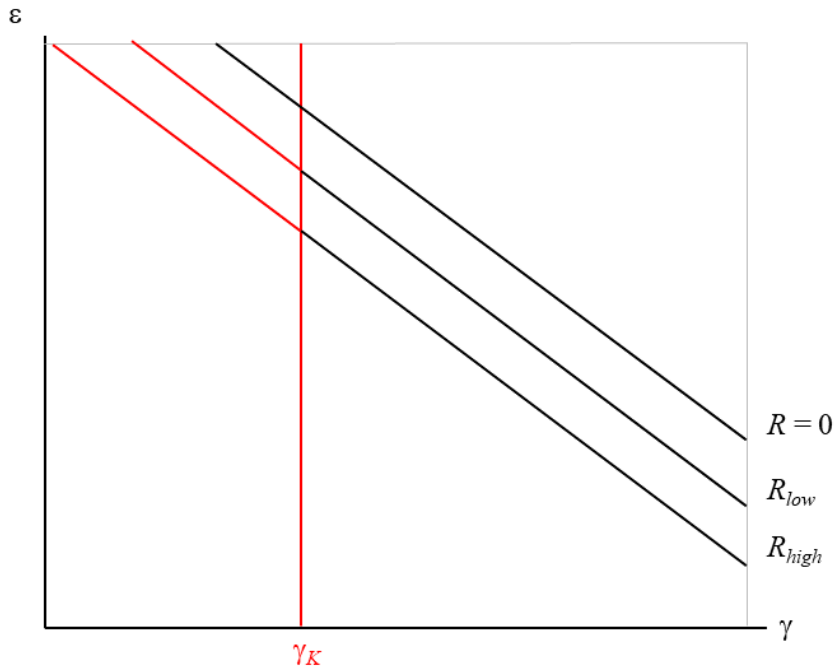


Figure 5: Solvency risk across reserve positions

Notes: The red vertical line denotes the solvency threshold from equation (19) and the black curve denotes the liquidity cutoff from equation (14). The red segment illustrates the insolvency region. Lower demand for reserves in the interbank market shifts this cutoff outward.

with the area below the threshold γ_R has increased while revenues stay the same. Thus, profits decrease. Banks respond by cutting on lending. Thus, as the policy rate increases, loan demand stays the same, but loan supply moves to the right (with curves depicted as in Figure 4 where lending rates are in the horizontal axis and quantities in the vertical axis). This is shown in Figure 7 for a change of the policy rate from 3 percent to 4 percent. This increase in lending rates generates a reduction in the amount of lending along the loan demand curve. Because deposit rates are kept constant at zero, the increase in the lending rate caused by the raise in policy rates automatically widens the net interest margin of banks as well as the deposit spread with respect to the official rate.

Figure 8 shows the effect of permanently keeping the official rate at different values between 1 percent and 10 percent. The top left Figure shows the spread between the lending rate, i^L, and the policy rate, i^O. We see the lending rate increases with the policy rate, making borrowing costlier for firms. Consequently, if we move to the right and from top to bottom in Figure 8, lending, L, decreases. Deposits, D, also fall. The reason is twofold. On the one hand, the creation of new deposits is smaller because lending has dropped. On the other hand, net worth of firms, N, which are maintained as deposits, also falls. The cut on lending also reduces bank leverage, L/K. The increase in the cost of borrowing together with the financial

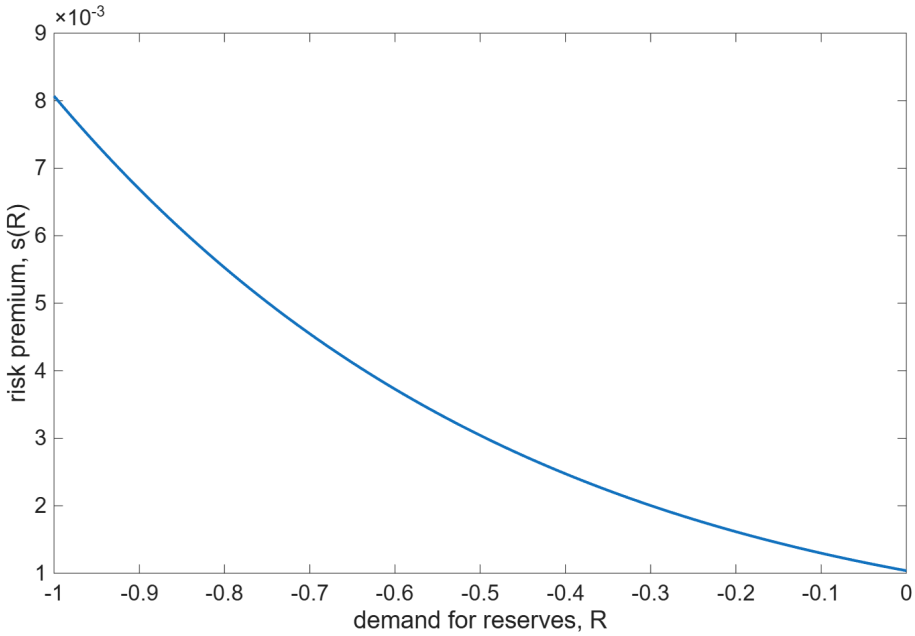


Figure 6: Risk premium in the interbank market

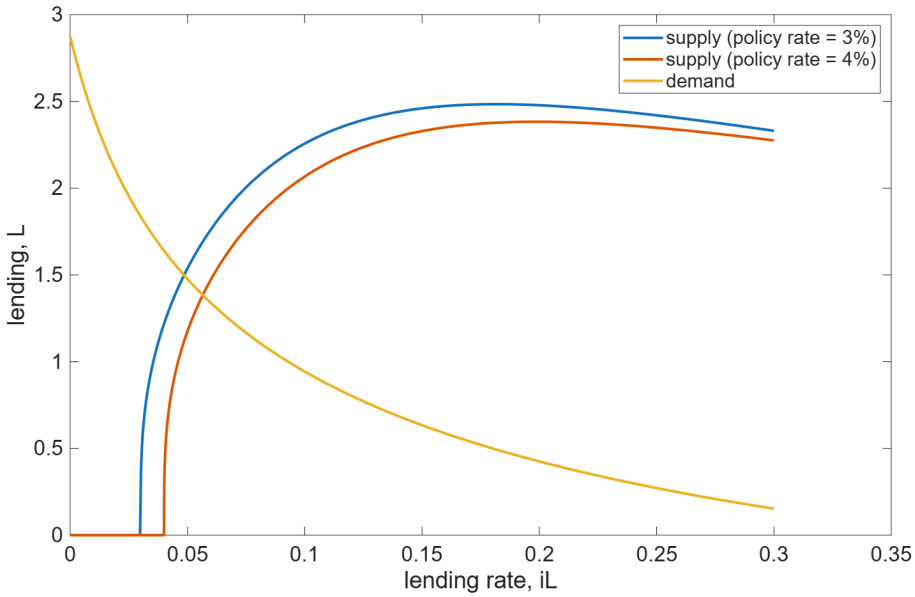


Figure 7: Effect of an increase in the policy rate on the lending market

repression generated by the hikes in official rates, make hiring more expensive which lowers the demand for labor depressing the economy with lower levels of output, y, and prices, W. This downturn is more pronounced the more contractionary monetary policy is. Furthermore, the last panel presents the NPL ratio of banks, $\Phi(\gamma_L)$, as well as the fraction of insolvent banks, $\Phi(\gamma_K)$. Interestingly, looser monetary policy, generated with lower policy rates, induces an increase in risk associated with these two margins. This is what the literature has labeled the risk-taking channel of monetary policy, [Borio and Zhu, 2012, Dell'Ariccia et al., 2017, Jiménez et al., 2014]. Importantly, in our economy, ex ante, all borrowers have the same risk. Thus, the larger exposure of banks as a response to monetary policy loosening is not due to banks reaching a pool of riskier borrowers. Instead, this increase in risk is just a consequence of the expansion

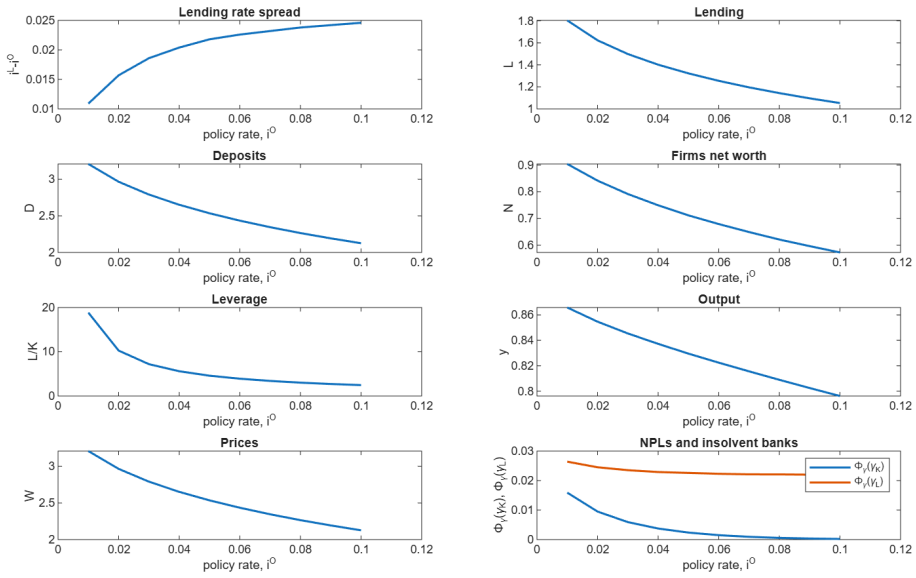


Figure 8: Equilibrium across policy rates

of credit by banks as the cost of funds decrease. The easing of monetary policy induces banks to expand credit compensating the reduction in costs with larger values for the thresholds γ_K and γ_L.

As Figure 8 shows, in our model, monetary policy has long run effects. That is, the economy sits on a different real equilibrium for each particular level of the official rate. Furthermore, in this economy, there are no transitions as we are assuming investors can reallocate funds between firms and banks instantly to equate the return on these investments. This was shown in the fourth panel of Figure 8. As the policy rate increases, the return on investing in bank capital rises above that of initial firm funds. Investors respond to this change by channeling less funds to firms and more to banks.

Two features deserve some words because they are not assumed but emerge from the mechanism. First, deposits, which are equal to the sum of lending, firm net worth and transfer to workers, $D = L + N + B$, fall as monetary policy becomes tighter, since the two endogenous components, L and N, decline, and they fall at the same time as the spread of the deposit rate over the policy rate widens. Notice that the deposit rate is fixed at zero while the official rate has risen. This is precisely the joint movement that Drechsler et al. [2017] identify as the deposit channel of the monetary policy. In our economy, however, the direction of causation is reversed since deposits fall because banks cut the supply of loans in response to a costlier liquidity environment, not because a deposit outflow forces banks to retrench. The same observable correlation is therefore consistent with a mechanism in which credit drives deposits rather than the reverse. Second, the rise in bank capital combined with the fall in lending drives bank leverage down (third line left), so that the model delivers a negative leverage response to a contractionary shock without any leverage constraint or capital regulation. Taken together, this shows that a single change in the cost of liquidity generates movements in credit, deposits, spreads, output, prices and leverage.

5.5 Balance sheet strength and the effects of monetary policy

Finally, Altunbaş et al. [2002], Ehrmann et al. [2001], Gambacorta and Mistrulli [2004], Gambacorta [2005], Kashyap and Stein [1995], Kishan and Opiela [2000, 2006], or van den Heuvel [2002]) study how the response of banks to changes in monetary policy depends on cross-sectional differences in their balance sheets. In particular, these papers suggest that banks cut back more on their lending in response to a contractionary monetary policy shock when they are less capitalized and/or more illiquid. For example, Gambacorta [2005] uses data for Italian banks and estimates a statistically significant drop in lending of 0.825 percent in response to a 1 percent increase in the policy rate. This drop in lending varies across the capitalization level of the bank. The average decrease is 0.622 percent for well capitalized banks while it is 0.976 percent for poorly capitalized banks. These effects remain statistically significant after the introduction of further controls and changes in the sample.

Figure 9 shows the percentage change drop in lending of an individual bank in response to an increase in the official rate i^O from 3 percent to 4 percent as a function of its capital. To produce that Figure we keep aggregate variables at the equilibrium levels and solve individual problems of banks imposing different capital levels ranging from 0.1 until 0.9. We see how the drop in lending in response to the increase in official rates is monotonically larger as banks maintain lower capital levels.

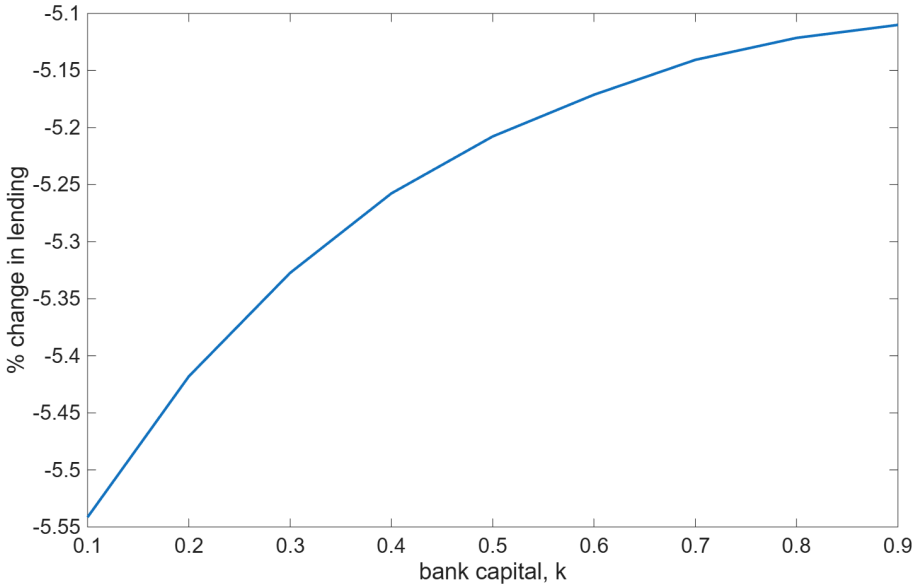


Figure 9: Changes in lending across capital levels

Capital affects the cost of liquidity faced by a bank through three channels, all of which are in Section 4. For given loans and reserves, more capital (i) moves the solvency threshold γ_K to the left, by $1/\mathbb{D}$ per unit of capital; (ii) moves the liquidity threshold $R = 0$ to the left by the same amount, since capital funds reserves; and (iii) lowers the premium that the bank pays at any given reserve position (Section 4.4.5). A bank with less capital is thus more exposed to illiquidity and insolvency, and its borrowing is priced as riskier, so that the cost of liquidity weighs more on its profits and a rise in the policy rate lowers its lending more.

6 Conclusions

This paper asks what limits banks' credit supply when neither a shortage of savings, nor a regulatory constraint, nor an agency problem does so, and what follows for bank risk and for monetary transmission. We have developed a general equilibrium model in which banks create deposits when they lend, and have shown that this single decision generates liquidity, credit, and solvency risk. First, the deposits created by a loan are spent and typically end up at other banks, so reserves must be transferred to settle the payment, which exposes the lending bank to liquidity risk. Second, borrowers can repay only if they attract enough deposits by selling their output; the more a bank lends, the larger the repayment burden and the greater the credit risk. Third, as non-performing loans rise, loan losses erode bank capital, which creates solvency risk. Because each of these exposures reduces expected profits, balance sheet expansion is bounded by an internal risk–profitability trade-off, and loan supply bends backward beyond a point even though information is symmetric between borrowers and lenders.

Calibrated to three moments of US data, the model delivers four further results, all of them untargeted. First, borrowing banks pay an interbank premium that increases with the amount borrowed and decreases with capital, because lenders cannot tell whether a deficit comes from a pure liquidity shock or from a shock that threatens solvency; this matches the direction of the evidence on the solvency/funding-cost nexus such as Aymanns et al. [2016], or Arnould et al. [2022]. Second, a higher policy rate raises the cost of liquidity and lowers the expected profit of expanding the balance sheet, so banks lend less; since lending creates deposits, deposits, output, and leverage fall too, with no leverage constraint, no capital regulation, and no deposit outflow forcing banks to retrench. The fall in lending is larger for less-capitalized banks, also in line with the direction of the evidence as in Gambacorta [2005]. Third, looser policy raises the share of insolvent banks and, to a lesser extent, the NPL ratio, which is a risk-taking channel [Borio and Zhu, 2012, Dell'Ariccia et al., 2017, Jiménez et al., 2014]; in our model it arises although all borrowers are ex ante identical, because lower funding costs lead banks to expand credit. Fourth, liquidity and credit risk are only weakly correlated at the bank level, as in Imbierowicz and Rauch [2014], even though for any given bank a larger loan volume raises all three risks together.

These results suggest a different reading of the deposits channel of monetary policy. A rise in the policy rate lowers deposits and widens the spread between the policy rate and the deposit rate, as in Drechsler et al. [2017], and lending falls. In our model the same co-movement appears, but causation runs from lending to deposits: deposits fall because banks create fewer of them, not because outflows force banks to cut lending. We have not shown that this mechanism dominates in the data; we have shown that the correlation alone cannot discriminate between the two.

Although we do not introduce regulation explicitly, the model could also offer new insights for the design of prudential and supervisory frameworks. Its message is that liquidity and solvency are linked because both stem from lending: interbank lenders read a large liquidity need as a signal of insolvency risk, and weaker capital raises a bank's exposure to illiquidity as well as to insolvency. Liquidity and capital requirements are set with different instruments, such as the liquidity coverage ratio and capital ratios, and the interaction between solvency

and liquidity has been examined empirically Pierret [2015], and Schmitz et al. [2019] as well as incorporated in proposals for joint stress tests [Bank for International Settlements, 2015, Cont et al., 2020, Gauthier et al., 2014]. Tarullo [2013] argues for tying capital requirements to the strength of a bank's liquidity position, since a bank that depends on short-term funding needs more equity to convince markets that it is solvent. Our results are consistent with the case for considering the two requirements together, and add that the link operates through the lending decision. We leave the evaluation of such joint regulations for future research.

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A The insolvency region

In this section, we show the conditions for the schedule $K' = 0$ to be to the left of the schedule $\gamma = \gamma_L$. In such a case, the schedule $K' = 0$ is formed by all combinations of γ and ϵ satisfying

$$K - i^O O + \gamma \mathbb{D} - L + \Pi^{IB} = 0. \tag{A.1}$$

We then ask where the schedule $K' = 0$ crosses the schedule $R = 0$ determined by (14). Notice, at that point, $\Pi^{IB} = 0$. For the crossing to be to the left of $\gamma = \gamma_L$, from (A.1) it must be the case that

$$\gamma = \frac{L - K + i^O O}{\mathbb{D}} < \frac{(1 + i^L)L}{\mathbb{D}} \equiv \gamma_L. \tag{A.2}$$

This will happen whenever $i^L L + K - i^O O > 0$. This condition is satisfied in the equilibrium of the model.

Second we ask whether the schedule $K' = 0$ crosses the schedule $\gamma = \gamma_L$. Given that the schedule $K' = 0$ has negative slope in the plane (γ, ϵ), this crossing should happen below the

schedule $R = 0$. In other words the crossing between $K' = 0$ and $\gamma = \gamma_L$ should happen at a value of ϵ satisfying

$$K - i^O O + i^L L - \gamma_L \mathbb{D} + [i^R + s(R)][O + K - L + (\gamma_L + \epsilon)\mathbb{D}] = 0, \tag{A.3}$$

or

$$\epsilon = \frac{(1 + i^O)O}{i^R + s(R)} - \frac{1 + i^R + s(R)}{i^R + s(R)}(\gamma_L - \gamma_R) \equiv \epsilon_L. \tag{A.4}$$

So, as long as the variable ϵ is bounded below by ϵ_L, and $i^L L + K - i^O O > 0$, the schedule $K' = 0$ will be to the left of the schedule $\gamma = \gamma_L$.

B Proof of Proposition 1

B.1 Preliminaries

By (13), (14), and (20), a lender who observes the position $\bar{R} < 0$ learns the sum $\gamma + \epsilon = \delta(\bar{R})$, where

$$\delta(\bar{R}) = \frac{\bar{R}}{\mathbb{D}} + \gamma_R, \qquad \gamma_R = \frac{L - K - O}{\mathbb{D}}. \tag{A.5}$$

The last expression follows from (14) with $A = N + K$ and $D = N + L + B$, since $D - B - A = L - K$. For a given value δ of the sum, define the joint density $f(x; \delta) \equiv \phi_\gamma(x)\, \phi_\epsilon(\delta - x)$ and, for any threshold c,

$$\Psi(c, \delta) \equiv \Pr(\gamma \geq c \mid \gamma + \epsilon = \delta) = \frac{\int_c^{\gamma_{\max}} f(x; \delta)\, dx}{\int_0^{\gamma_{\max}} f(x; \delta)\, dx},$$

which is (21) with $c = \gamma_K$ and $\delta = \delta(\bar{R})$ (and equals one if $c \leq 0$). By (22),

$$s(\bar{R}) = (1 + i^R)\, \frac{1 - \Psi}{\Psi}, \tag{A.6}$$

which is a strictly decreasing function of Ψ on $(0, 1]$. It therefore suffices to show that $\Psi(\gamma_K, \delta(\bar{R}))$ is non-increasing in $|\bar{R}|$ and non-decreasing in K.

B.2 Two lemmas

Lemma 1: Log-concavity implies the monotone likelihood ratio property. Let g be a log-concave density and let $\delta' > \delta$. Then for all $x_1 < x_2$,

$$g(\delta' - x_2)\, g(\delta - x_1) \;\geq\; g(\delta' - x_1)\, g(\delta - x_2),$$

that is, $g(\delta' - x)/g(\delta - x)$ is non-decreasing in x wherever both are positive.

Proof. Let $a = \delta - x_2$ and $b = \delta' - x_1$. Then $a < \delta - x_1 < b$ and $a < \delta' - x_2 < b$, and $(\delta - x_1) + (\delta' - x_2) = a + b$. Let $h = \ln g$, which is concave. For a concave function and points $u, v \in [a, b]$ with $u + v = a + b$, we have $h(u) + h(v) \geq h(a) + h(b)$, because u and v are convex combinations of a and b: $u = \lambda a + (1 - \lambda)b$ and $v = (1 - \lambda)a + \lambda b$ with the same λ, so

$h(u)+h(v) \geq \lambda h(a)+(1-\lambda)h(b)+(1-\lambda)h(a)+\lambda h(b) = h(a)+h(b)$. Taking $u = \delta - x_1$ and $v = \delta' - x_2$ gives the inequality after exponentiating.

Lemma 2: Stochastic dominance of posteriors. Let $f_1, f_2 \geq 0$ be integrable functions on an interval I with positive integrals, such that $f_2(x)f_1(y) \geq f_1(x)f_2(y)$ whenever $x \geq y$. Then for every c, $\int_{x\geq c} f_2 / \int_I f_2 \;\geq\; \int_{x\geq c} f_1 / \int_I f_1$.

Proof. Write $U_i = \int_{x\geq c} f_i$ and $V_i = \int_{x<c} f_i$. The claim $U_2/(U_2+V_2) \geq U_1/(U_1+V_1)$ is equivalent to $U_2V_1 \geq U_1V_2$. But

$$U_2V_1 - U_1V_2 = \iint_{x\geq c>y} \big[f_2(x)f_1(y) - f_1(x)f_2(y)\big]\,dx\,dy \;\geq\; 0,$$

since $x > y$ in the region of integration.

B.3 Proof of part (i): the amount borrowed

Fix L, O, K, and $\mathbb{D}$, so that γ_K and γ_R are fixed. Let $\bar{R}' < \bar{R} < 0$, so that $|\bar{R}'| > |\bar{R}|$. By (A.5), $\delta' \equiv \delta(\bar{R}') < \delta \equiv \delta(\bar{R})$. Take $f_1(x) = f(x;\delta')$ and $f_2(x) = f(x;\delta)$. By Lemma 1 with $g = \phi_\epsilon$ (log-concave), $f_2(x)f_1(y) \geq f_1(x)f_2(y)$ for $x \geq y$. Lemma 2 with $c = \gamma_K$ gives

$$\Psi(\gamma_K, \delta) \;\geq\; \Psi(\gamma_K, \delta').$$

Hence Ψ is non-increasing in $|\bar{R}|$ and, by (A.6), s is non-decreasing in $|\bar{R}|$. If ϕ_ϵ is strictly log-concave, the inequality in Lemma 1 is strict for $x_1 < x_2$, the double integral in Lemma 2 is positive whenever both sides of c have positive mass, and the monotonicity is strict. □

B.4 Proof of part (ii): capital

Fix L, O, $\mathbb{D}$, and the position $\bar{R}$, and consider a rise in capital from K to $K+\Delta K$, with $\Delta K > 0$ and $c \equiv \Delta K/\mathbb{D}$. Three terms depend on K:

$$\gamma_K = \frac{L - K + i^O O}{\mathbb{D}} \;\longrightarrow\; \gamma_K - c,$$

$$\gamma_R = \frac{L - K - O}{\mathbb{D}} \;\longrightarrow\; \gamma_R - c,$$

$$\delta(\bar{R}) \;\longrightarrow\; \delta(\bar{R}) - c.$$

Write $\delta = \delta(\bar{R})$, so that the new inferred sum is $\delta - c$. Since $\gamma = \delta - \epsilon$ given the sum δ, the event $\{\gamma \geq \gamma_K\}$ is the event $\{\epsilon \leq \delta - \gamma_K\}$. Therefore

$$\begin{aligned}
\Psi_{\text{old}} &= \Pr\big(\epsilon \leq \delta - \gamma_K \mid \gamma + \epsilon = \delta\big),\\
\Psi_{\text{new}} &= \Pr\big(\epsilon \leq (\delta - c) - (\gamma_K - c) \mid \gamma + \epsilon = \delta - c\big)\\
&= \Pr\big(\epsilon \leq \delta - \gamma_K \mid \gamma + \epsilon = \delta - c\big).
\end{aligned}$$

The cut-off $\delta - \gamma_K$ is the same in both expressions; only the conditioning value of the sum changes, from δ to $\delta - c < \delta$. The conditional density of ϵ given $\gamma + \epsilon = \delta$ is proportional to

$e \mapsto \phi_\epsilon(e)\,\phi_\gamma(\delta - e)$. Apply Lemma 1 with $g = \phi_\gamma$ (log-concave) and the variable e in place of x: for $\delta' > \delta''$, $\phi_\gamma(\delta' - e)/\phi_\gamma(\delta'' - e)$ is non-decreasing in e, so the conditional density of ϵ given the sum δ' dominates that given δ'' in the monotone likelihood ratio order. By Lemma 2, the conditional distribution of ϵ given a lower sum is stochastically smaller, so $\Pr(\epsilon \le y \mid \delta - c) \ge \Pr(\epsilon \le y \mid \delta)$ for every y. Hence $\Psi_{\text{new}} \ge \Psi_{\text{old}}$, and by (A.6) the premium is non-increasing in K; it is strictly decreasing under strict log-concavity. □

C Derivation of expression (24) and (25)

Expected bank capital is equal to

$$\begin{aligned}
E(K') &= \int_{\gamma_K}^{\gamma_{\max}} \int_0^{\epsilon_{\max}} K' \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) \\
&= \int_{\gamma_K}^{\gamma_L} \int_0^{\gamma_R - \gamma} \left[K - i^O O + \gamma\mathbb{D} - L + (i^R + s(R))R\right] \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) \\
&+ \int_{\gamma_K}^{\gamma_L} \int_{\gamma_R - \gamma}^{\epsilon_{\max}} \left[K - i^O O + \gamma\mathbb{D} - L + i^R R\right] \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) \\
&+ \int_{\gamma_L}^{\gamma_{\max}} \int_0^{\gamma_R - \gamma} \left[K - i^O O + i^L L - i^D \gamma\mathbb{D} + (i^R + s(R))R\right] \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) \\
&+ \int_{\gamma_L}^{\gamma_{\max}} \int_{\gamma_R - \gamma}^{\epsilon_{\max}} \left[K - i^O O + i^L L - i^D \gamma\mathbb{D} + i^R R\right] \Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma),
\end{aligned}$$

where the interbank rate charged or received depends on whether reserves are positive, so that $\epsilon < \gamma_R - \gamma$ or negative, so that $\epsilon > \gamma_R - \gamma$. Applying the definition of γ_L, γ_K, and $G(\underline{\gamma})$ in (15), (19), and (23), respectively, yields (24) and (25).

D Derivation of expressions (26) and (27)

First, taking first order condition with respect to O in (24) yields

$$\frac{\partial E(K')}{\partial O} = \mathbb{D}\frac{dG(\gamma_K)}{d\gamma_K} \times \frac{\partial \gamma_K}{\partial O} + \frac{\partial \Pi(O, L)}{\partial O}. \tag{A.7}$$

Furthermore, notice

$$\frac{dG(\underline{\gamma})}{d\underline{\gamma}} = -\left[1 - \Phi_\gamma(\underline{\gamma})\right],$$

$$\frac{\partial \gamma_K}{\partial O} = \frac{i^O}{\mathbb{D}},$$

and,

$$\frac{\partial \Pi(O, L)}{\partial O} = i^R\left[1 - \Phi_\gamma(\gamma_K)\right] + \int_{\gamma_K}^{\gamma_{max}} \int_{\epsilon_{min}}^{\gamma_R - \gamma} s(R)\Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) + o.t.O$$

where $o.t.O$ are other terms involving the differentiation of the limits of the integrals in $\Pi(O, L)$. These terms are in general small and approximated to zero. Also, there is the change in $s(R)$ associated with changes in R produced by the choice of O. We assume banks take $s(R)$ as given so this term is also zero. Collecting all these terms in (A.7) provides (26).

On the other hand, the first order condition with respect to L in (24) yields

$$\frac{\partial E(K')}{\partial L} = \mathbb{D}\frac{dG(\gamma_K)}{d\gamma_K} \times \frac{\partial \gamma_K}{\partial L} - \mathbb{D}\frac{dG(\gamma_L)}{d\gamma_L} \times \frac{\partial \gamma_L}{\partial L} + \frac{\partial \Pi(O,L)}{\partial L}. \tag{A.8}$$

Then, notice

$$\frac{\partial \gamma_K}{\partial L} = \frac{1}{\mathbb{D}},$$
$$\frac{\partial \gamma_L}{\partial L} = \frac{1+i^L}{\mathbb{D}},$$

and,

$$\frac{\partial \Pi(O,L)}{\partial L} = -i^R\left[1-\Phi_\gamma(\gamma_K)\right] - \int_{\gamma_K}^{\gamma_{max}}\int_{\epsilon_{min}}^{\gamma_R-\gamma} s(R)\Phi_\epsilon(d\epsilon)\Phi_\gamma(d\gamma) + o.t.L$$

where, again, $o.t.L$ involves terms associated with the differentiation of the limits of the integrals in $\Pi(O,L)$ as well as the change in $s(R)$, and approximated to zero. Consolidating all these terms in (A.7) and using (26), provides (27).

E Proof of Proposition 2

E.1 Derivatives with respect to L

Using $A = N + K$ and $D = N + L + B$ in (14), and holding fixed $\mathbb{D}$, O, K, N, B, and i^L,

$$\gamma_R = \frac{L-K-O}{\mathbb{D}}, \qquad \gamma_L = \frac{(1+i^L)L}{\mathbb{D}}, \qquad \gamma_K = \frac{L-K+i^O O}{\mathbb{D}},$$

so that $\partial\gamma_R/\partial L = \partial\gamma_K/\partial L = 1/\mathbb{D}$ and $\partial\gamma_L/\partial L = (1+i^L)/\mathbb{D}$, all positive because $\mathbb{D} > 0$ and $i^L > -1$. Thus, a larger loan volume raises the liquidity, credit, and solvency thresholds, that is, all three exposures. □

E.2 The differences (28)–(30)

Subtracting the expressions above,

$$\gamma_K - \gamma_R = \frac{(L-K+i^O O)-(L-K-O)}{\mathbb{D}} = \frac{(1+i^O)\,O}{\mathbb{D}},$$
$$\gamma_L - \gamma_R = \frac{(1+i^L)L-(L-K-O)}{\mathbb{D}} = \frac{i^L L+K+O}{\mathbb{D}},$$
$$\gamma_L - \gamma_K = \frac{(1+i^L)L-(L-K+i^O O)}{\mathbb{D}} = \frac{i^L L+K-i^O O}{\mathbb{D}},$$

which are (28), (29), and (30). □

E.3 The symmetric equilibrium

Interbank-market clearing requires $\int_0^1 R(j)\,dj = 0$ (equation (37)). By (13),

$$\int_0^1 R(j)\,dj = O + B + A + \mathbb{D}\int_0^1 (\gamma_j + \epsilon_j)\,dj - \int_0^1 D_j\,dj.$$

By the law of large numbers across nodes, $\int_0^1 \gamma_j \, dj = E(\gamma) = 1$ and $\int_0^1 \epsilon_j \, dj = E(\epsilon) = 0$. In a symmetric equilibrium all banks choose the same L, so $D_j = D$ and aggregate payments equal each bank's deposits, $\mathbb{D} = D$. Hence $\int_0^1 R(j) \, dj = O + B + A + \mathbb{D} - D = O + B + A$, and market clearing implies

$$O = -(A + B).$$

Then, by (14) and $A = N + K$,

$$\gamma_R = \frac{L - K - O}{\mathbb{D}} = \frac{L - K + A + B}{\mathbb{D}} = \frac{L + N + B}{D} = 1,$$

since $D = N + L + B = \mathbb{D}$. □

E.4 The ordering $\gamma_K < \gamma_L < \gamma_R$

Take $i^O \geq 0$ and $i^L \geq 0$, as in the calibration, and $K > 0$. We examine the three differences in (28)-(30) at $O = -(A + B)$.

- By (28), $\gamma_K - \gamma_R = -(1 + i^O)(A + B)/\mathbb{D} < 0$, so $\gamma_K < \gamma_R$ always.
- By (30), $\gamma_L - \gamma_K = \left[i^L L + K + i^O(A + B)\right]/\mathbb{D} > 0$, so $\gamma_K < \gamma_L$ always.
- Finally, $\gamma_L < \gamma_R = 1$ if and only if $(1 + i^L)L < \mathbb{D} = N + L + B$, that is, if and only if $i^L L < N + B$. (Equivalently, by (29), $\gamma_L - \gamma_R = [i^L L + K - (A + B)]/\mathbb{D} = [i^L L - N - B]/\mathbb{D}$, since $A = N + K$.)

Therefore the chain $\gamma_K < \gamma_L < \gamma_R$ holds if and only if $i^L L < N + B$. □

F Derivation of expression (33)

By the definition of γ_K, $\mathbb{D}(1 - \gamma_K) = \mathbb{D} - L + K - i^O O = A + B - i^O O$, using $\mathbb{D} = N + L + B$ and $A = N + K$. Since $E(\gamma) = 1$, $G(\gamma_K) - E[(\gamma_K - \gamma)^+] = E(\gamma - \gamma_K) = 1 - \gamma_K$, so that

$$\mathbb{D}\, G(\gamma_K) = \mathbb{D}(1 - \gamma_K) + \mathbb{D}\, E[(\gamma_K - \gamma)^+] = A + B - i^O O + \Sigma,$$

where $\Sigma = \mathbb{D}\, E[(\gamma_K - \gamma)^+]$ is the expected negative part of (17). Adding $\mathcal{N}' = \mathbb{D}G(\gamma_L)$ and $\mathcal{K}'$ from (32) gives (33). □